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A comprehensive review of single-phase immersion cooling in data centres

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A R T I C L E I N F O

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A B S T R A C T

In recent years, the rapid surge in global computing power demand has posed significant challenges for data centers in terms of chip-level and rack-level cooling, as well as system-level energy efficiency. With chips exceeding 700 W and racks surpassing 30 kW, liquid cooling has become the preferred solution. Among various liquid cooling methods, single-phase immersion cooling, which utilizes high-boiling-point dielectric fluids to cool servers, offers distinct advantages such as high heat dissipation efficiency, excellent stability, low cooling energy consumption, and minimal noise. These attributes make it a highly promising solution for addressing the thermal management challenges in high-performance data centers, garnering widespread attention from both academia and industry. However, the relatively poor thermal properties of immersion coolants, combined with the emergence of kW-level digital chips, have challenged the cooling performance of conventional single-phase immersion cooling. In the above context, this paper analyzes and compares the thermo-physical properties, material compatibility and signal integrity of various single-phase immersion coolants, systematically reviews heat dissipation enhancement techniques and structural optimization studies at the chip, server and cabinet level, and discusses the energy saving advantages and potential of single-phase immersion cooling in data centers. This work aims to provide a systematic report on single-phase immersion cooling research, assess its technological development trends, and further explore its cooling potential. Finally, we summarize the limitations of current studies and analyze potential future research directions and applications, with a particular focus on the technical pathways for achieving chip cooling with power dissipation exceeding 1000 W and 100 W/cm^2 using single-phase immersion cooling.

1. Introduction

The rapid advancement of digital technologies, such as artificial intelligence, big data, cloud computing, and 5G [1,2], has significantly increased the demand for data centers. As hubs for data storage, network communication, and computation, data centers are growing rapidly in both number and scale. However, this explosive expansion comes at the cost of increased energy consumption and thermal management challenges. In 2018, the total energy consumption of data centers was approximately 205 TWh, accounting for about 1 % of global energy consumption [3]. This proportion is expected to reach 1.86 % by 2030 [4]. As Fig. 1(a) shows, in a typical data center, over 40 % of the energy is consumed by thermal management systems [5], leading to significant energy waste. Meanwhile, the continuous increase in transistor density has led to a higher heat dissipation per unit area of the chip, further intensifying the challenges of thermal management. CPU power consumption in modern data centers exceeds 300 W [6], with advanced GPUs consuming up to 1000 W [7]. At the rack level, the global average rack power density increased linearly by 3.5 times from 2010 to 2020. A survey conducted by the Uptime Institute in 2024 revealed that, among the 721 data centers surveyed, 17 % of them had a maximum rack power greater than 30 kW [8]. Effectively cooling high-heat-flux chips and high-power-density racks has become a critical challenge for data centers. The failure rate of electronic devices increases sharply with rising temperatures [9]. As shown in Fig. 1(b), studies have shown that more than half of integrated circuit failures are caused by thermal issues [10,11]. If the heat generated by servers is not promptly dissipated, it can lead to server overheating and potential failures [12]. Balancing energy consumption and cooling performance, the development of more cost-effective and efficient cooling technologies is essential for ensuring the efficient and reliable operation of future data centres.

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Nomenclature

Symbols

C_p	specific heat capacity, J/kg·K
H	fin high, mm
L	length, mm
Nu	Nusselt number
P	power consumption, W
p	fin pitch, mm
Q	heat load for single chip, W
Q_{total}	maximum heat loads of cooling system, W
q	heat flux, W/cm^2
R	Resistance, K/W
Re	Reynold number
T	temperature, ℃
t	fin thickness, mm
V	volume flow rate, L/min
v	flow velocity, m/s
W	width, mm
λ	thermal conductivity of fluid, W/(m·K)
μ	dynamic viscosity, mPa·s
ρ	density, kg/m^3

Abbreviations

AI	Artificial Intelligence
CRAC	Computer Room Air Conditioner
CPU	Central Processing Unit
D_k	Dielectric Constant
ERF	Energy Reuse Factor
FOM	Figure of Merit
GPU	Graphics Processing Unit
GWP	Global Warming Potential
IC	Immersion Cooling
PCB	Printed Circuit Board
PUE	Power Usage Effectiveness
pPUE	partial Power Usage Effectiveness
PVC	Polyvinyl Chloride
SPIC	Single-phase immersion cooling
VC	Vapor Chamber
WC	water-cooling

Subscript

bf	heat sink base to fluid
cf	case to fluid
cooling	cooling system
f	fluid
ICT	entire data center
IT	Information Technology equipment
in	inlet
jf	junction to fluid
max	maximum
other	other infrastructure
out	outlet
reuse	energy reused
w	cooling water

Traditional air-cooling methods are increasingly considered inadequate for meeting the thermal management requirements of high-heat-flux electronic devices [13], and the energy consumption associated with compressor-based refrigeration cycles is substantial [14]. When the thermal design power (TDP) of a chip exceeds 250–280 W, the low heat transfer coefficient of air often results in localized hotspots, making it difficult to maintain the chip temperature within an acceptable range [15,16]. Consequently, the maximum rack power in traditional air-cooled data centers is typically limited to 20–30 kW [17]. As chip and rack thermal power continues to rise, the inherent thermal-physical limitations of air make effective cooling increasingly challenging, suggesting that liquid coolants may serve as a more suitable alternative to air [18].

Data center liquid cooling can be broadly categorized into two main approaches: indirect liquid cooling (e.g., cold plates [19,20], heat pipes [21,22], and thermosyphons [23,24]) and direct liquid cooling (e.g., immersion cooling, jet impingement [25], and spray cooling [26]). Direct immersion cooling technology operates by fully submerging heat-generating components in a coolant, leveraging liquid convection heat transfer to absorb the heat produced by servers and other equipment. This absorbed heat is then transferred to a cooling water loop via a heat exchanger, ultimately dissipating into the environment [27]. Direct immersion liquid cooling can be further classified into single-phase and two-phase types. While two-phase immersion cooling can effectively cool high heat flux chips by utilizing the latent heat of the coolant [28–31], its flow and heat transfer processes are complex, and the phase change induces significant pressure variations, leading to stability issues in practical applications [12]. Additionally, it entails higher investment

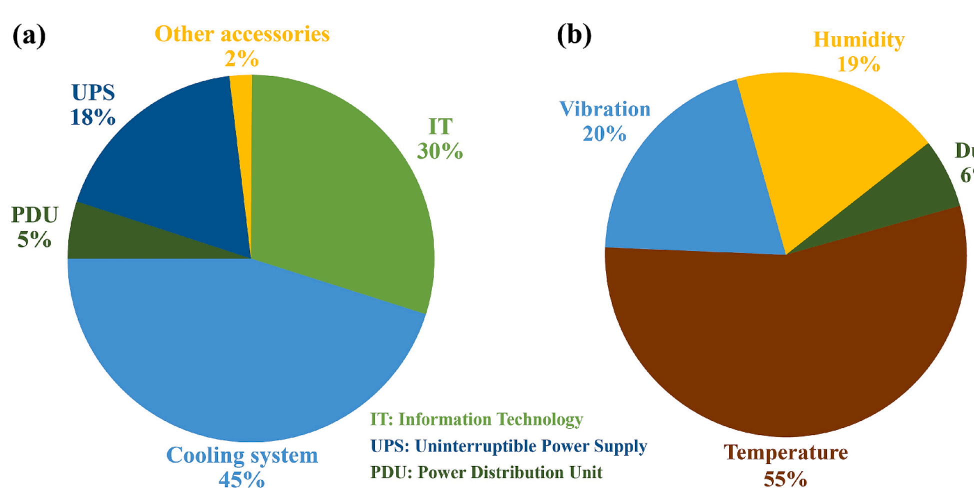


Fig. 1. Schematic of (a) Typical power distribution in data centers [5], (b) Failure Cause Distribution of Electronic Devices [11].

and maintenance costs [32]. In contrast, single-phase immersion cooling (SPIC) has a clearer mechanism, also offers high thermal performance, and is more suitable for commercialization in data center applications.

Fig. 2 compares the advantages and disadvantages of various cooling technologies based on commonly used engineering evaluation metrics. Currently, cold plate liquid cooling is the most widely adopted liquid cooling technology in data centers [33]. In terms of thermal performance, cold plate cooling benefits from the superior thermophysical properties of the coolant compared to SPIC [34]. However, SPIC reduces solid thermal resistance [35], enabling higher heat dissipation efficiency and simultaneous efficient cooling of memory and other components. The numerical simulations by Kim et al. [36] demonstrated that under SPIC conditions, not only was effective cooling achieved for an 800 W CPU, but the temperature of a 20 W TDP memory modules was also maintained at just 44.2 °C, significantly lower than that of air cooling (113.7 °C) and hybrid cooling combining cold plate and air cooling (86.2 °C). Furthermore, cold plate liquid cooling with straight channels tends to exhibit non-uniform temperature distribution along the upstream and downstream sections when cooling high-power chips [37], whereas the higher total specific heat capacity of immersion coolants facilitates a more uniform chip temperature distribution [18]. Regarding investment costs and energy consumption, cold plate liquid cooling is highly compatible with existing data center infrastructure, requiring minimal modifications to server architecture and enabling a cost-effective transition from air-cooled to liquid-cooled data centers. In contrast, SPIC data centers necessitate stringent compatibility analysis for the coolant and involve higher initial investment costs [38]. However, SPIC systems eliminate the need for air cooling components. Compared to cold plate systems, where approximately 30 % of the heat still needs to be dissipated by air cooling [39], SPIC eliminates the need for high-energy, low-efficiency components such as fans, chilled water units, and CRAC units. This results in lower cooling system energy consumption and a more simplified data center layout [40,41]. Additionally, SPIC systems offer advantages such as lower noise levels, simplified assembly, and the elimination of indirect cooling leakage risks, significantly reducing maintenance costs in data centers [12,42,43]. Moreover, compared to jet impingement and spray cooling, immersion cooling exhibits notable advantages in system complexity, long-term reliability, and energy efficiency [38]. Therefore, SPIC technology represents the most effective solution currently available for the efficient cooling of high heat flux electronic devices and high-power-density servers and cabinets.

In terms of cooling capacity, SPIC can achieve up to ten times the cooling performance of forced air cooling [44], with a single cabinet cooling capacity exceeding 300 kW [45]. Regarding power consumption, CPU temperature is directly related to its power consumption, and a lower CPU temperature can significantly reduce power usage [46]. The adoption of SPIC technology can reduce data center cooling power consumption by 95 % and server power consumption by 10–20 % [47]. In terms of reliability, immersion cooling isolates the PCB from direct contact with air, protecting it from airborne contaminants and minimizing the impact of ambient humidity on operational safety [48]. Compared to air-cooled servers, SPIC servers exhibit an approximately 50 % reduction in the average failure rate of components [49]. Moreover, SPIC systems demonstrate reliability even under extreme conditions while maintaining high cooling performance [50]. Given its promising application prospects in electronics thermal management, many companies have begun actively deploying immersion cooling solutions, and the adoption of SPIC technology in data centers is expected to see significant advancements in the future [51].

Data center SPIC systems can be categorized into two operating modes based on the convection heat transfer mechanism: natural convection and forced convection, as shown in Fig. 3. In the single-phase natural convection immersion cooling mode, the coolant is not externally driven; its flow is induced by density differences resulting from temperature variations. Heat is typically removed from the system to the external environment via cooling coils inside the immersion tank. This mode is also referred to as buoyancy-driven immersion cooling [52]. In the single-phase forced convection immersion cooling mode, the coolant is circulated by a pump, maintaining a certain flow velocity within the immersion tank. Heat exchange occurs between the coolant and the cooling water via a heat exchanger located outside the cabinet. This mode, offering better convective heat transfer performance, is also known as pump-driven immersion cooling [53]. In forced convection systems, the cooling modes can generally be classified into three categories based on the internal flow characteristics of the server. The first

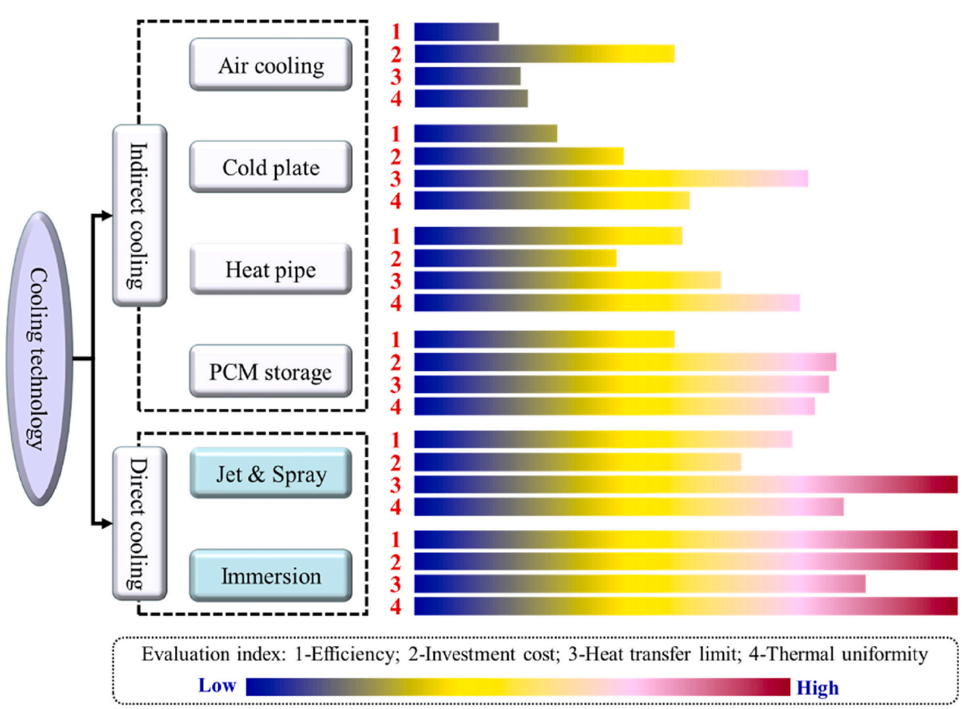


Fig. 2. Evaluation of electronic device cooling technologies [38].

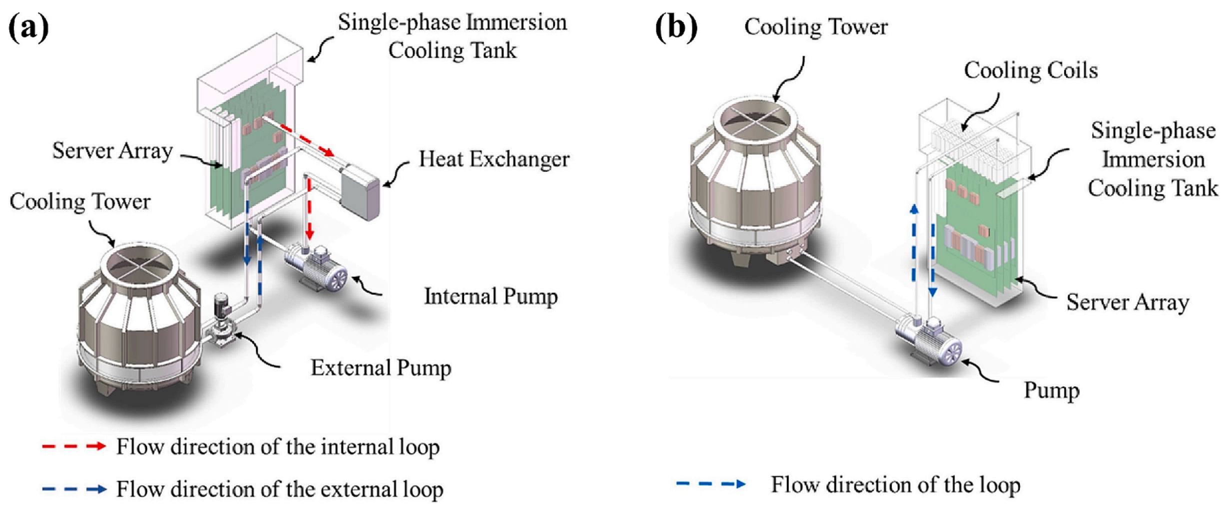


Fig. 3. SPIC model: (a) Forced convection immersion cooling, (b) Natural convection immersion cooling [53].

category is the constrained flow mode, where the flow conditions are influenced solely by the inlets, outlets, and the internal structure of the server. The second category is the internally driven flow mode, where the fluid flow is regulated by an internal power source (such as a micro-pump or propeller), resulting in localized changes in the flow field. The third category is the direct-to-chip cooling mode, in which the coolant inlet is located directly in the high heat flux density chip area. These three modes can all improve heat dissipation efficiency by optimizing the heat sink and the internal structure of the server. Existing research on cabinet-level structural optimization mainly focuses on the first two cooling modes, with a lack of cabinet-level research on the third mode.

Based on the thermophysical properties of liquids, both SPIC methods exhibit higher cooling performance compared to conventional forced air cooling methods, capable of reducing the peak temperature of devices by more than 20 % [54,55]. Compared to natural convection, forced convection immersion liquid cooling method has a faster overall flow velocity of the coolant, which can achieve better heat dissipation performance. Due to the superior heat dissipation performance of pump-driven immersion cooling, it can achieve the same heat dissipation performance as natural convection at a higher coolant temperature. As a result, the secondary side energy consumption of the system may be lower, which suggests that its energy consumption level could be more efficient than that of buoyancy-driven immersion cooling methods [53]. Therefore, it is worthwhile to evaluate and analyze the heat dissipation capacity and energy consumption levels of the system to construct a data center single-phase immersion liquid cooling system with high heat dissipation performance and a lower energy consumption level.

As shown in Fig. 4, the maximum heat flux of chips in data centers and the power density of the 42U high-power racks have increased rapidly, especially in recent years with the advancement of AI technologies, nearly exhibiting an exponential growth. However, most research on single-phase immersion cooling is still focused on medium- to low-heat flux levels, and there is a lack of systematic summaries specifically addressing SPIC research for data centers. This paper provides a systematic literature review summarizing the current research and application status of single-phase immersion cooling in data centers. The focus is on the selection of immersion cooling fluids and the challenges faced by data centers in chip-level, server-level, cabinet-level cooling, and system-level energy saving and consumption reduction. The review covers research progress on single-phase immersion cooling in data centers since 2013. Through this analysis, we evaluate whether single-phase immersion cooling technology can meet the cooling demands of data centers under high heat flux conditions in the future. Additionally, we predict the development trends of single-phase immersion cooling technology in data centers, summarize existing research gaps, and analyze possible future research directions and application prospects. This will provide an important reference for the future development of single-phase immersion cooling technology in data centers.

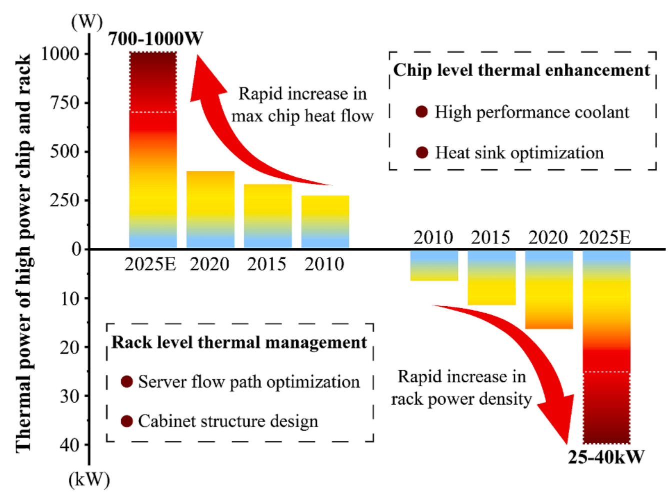


Fig. 4. Trend of chip and high-power rack heat flux increase.

2. Single phase immersion cooling liquid

Currently, the cooling media commonly used in immersion cooling systems for electronic equipment in data centers primarily include electrically insulating fluorinated electronic liquids and oil coolants. There are also studies exploring immersion cooling using water as the working fluid by applying insulating coatings [56,57], however the reliability of these coatings requires further investigation. The thermal performance of the immersion coolant largely affects the heat dissipation ability of the SPIC system due to the huge difference in thermal performance parameters such as viscosity, thermal conductivity, and specific heat capacity of different types of coolant. Therefore, it is necessary to analyze the advantages and disadvantages of their respective heat dissipation as well as the conditions of application. Since the coolant will directly contact the electronic components inside the server, material compatibility and electrical properties must also be considered. This section discusses the thermo-physical properties of single-phase immersion coolants for data centers, material compatibility with various components within the server, stability, environmental friendliness, and cost. In addition, statistics and analysis of nanofluid research in data centers are presented.

2.1. Electronic fluorinated liquids

The electronic fluorinated liquids used in SPIC primarily include hydrofluoroethers (HFEs) and perfluorocarbons (PFCs). HFEs are environmentally friendly coolants with ozone-friendly properties and low Global Warming Potential (GWP) [58]. But PFCs are potent greenhouse gases [59], with a GWP typically exceeding several thousand [60].

One significant advantage of using fluorinated liquids as immersion cooling media is their markedly lower viscosity compared to traditional oil coolants. Viscosity plays a crucial role in determining both heat dissipation performance and pressure drop. Numerous studies have shown that among similar coolants, viscosity has a more pronounced impact on flow and heat transfer performance than other thermophysical properties [61,62]. Chen et al. [62] found that using low-viscosity fluorinated liquids compared to high-viscosity ones reduced the CPU's maximum temperature by over 12 % and decreased pressure loss by more than 40 %. An increase in viscosity thickens the boundary layer during fluid flow, reduces near-wall flow velocity, and consequently lowers the Reynolds number (Re) and Nusselt number (Nu). This results in a decline in heat transfer performance and worsened temperature uniformity.

Fluorinated liquids were initially used for cooling and cleaning electronic devices due to their high inertness, excellent stability, and low reactivity with other substances, making them highly compatible with various materials inside servers [38]. Research and testing conducted by the Open Compute Project have shown that PFCs exhibit high compatibility with most materials commonly used in current server architectures [63]. Liu et al. [64] experimentally investigated the compatibility between the fluorinated liquid FC-40 and a commonly used thermal interface material, PTM795X. The results indicated good compatibility, with a weight change of less than 1 % after 300 h of immersion and no degradation in thermal conductivity. Further supporting this, Wang et al. [65] operated hundreds of servers continuously in fluorinated liquid for six months without observing any compatibility issues. Additionally, measurement data from Alibaba Cloud's single-phase immersion server clusters operating for over four years reveal that the performance indicators of servers immersed in electronic fluorinated liquids meet all requirements. Furthermore, no significant changes in electrical or physical properties were observed, and the failure rate of IT equipment was lower than that of air-cooled systems, demonstrating enhanced system reliability [49]. In summary, fluorinated liquids have demonstrated outstanding material compatibility in immersion cooling applications, with minimal reported issues. This exceptional compatibility provides a strong foundation for the operational reliability of SPIC systems.

Signal compatibility is a critical electrical characteristic of immersion cooling liquids, primarily concerning their impact on the signal integrity of electronic devices [38]. The dielectric constant (D_k) serves as a key parameter in evaluating this property. Kurata et al. [66] investigated the electrical performance of silicon photonic micro-transceivers operating at a high-frequency signal of 25.78 Gbps in different fluorinated liquids. Their results showed that when the cooling liquid had a dielectric constant of 1.2, the bit error rate (BER) was as low as 10^{-12}. However, in a cooling liquid with a dielectric constant of 7.5, signal integrity was significantly degraded. This finding confirms that immersion cooling liquids with dielectric constants below 2 exhibit excellent signal compatibility. Liu et al. [67] examined the performance of high-speed interconnects (64 Gbps, PAM4) under air cooling and immersion cooling conditions (Dk = 1.9, 2.1). Their findings indicated that the primary factors affecting signal integrity were impedance reduction and increased reflection noise due to the higher dielectric constant. Based on this, they suggested that interconnect systems require specialized design optimizations in immersion cooling environments. Further, Liu et al. [68] evaluated the performance of 100G-PAM4 high-speed Ethernet links under both air and immersion cooling (Dk ≈ 2). Compared to air, immersion in a cooling liquid with Dk = 2 led to a reduction in connector impedance from 90 Ω to 70 Ω and an increase in BER by two orders of magnitude. This study underscores the necessity of optimizing connector designs to adapt to immersion cooling environments.

2.2. Oil coolants

Because of the high cost, volatility, and negative environmental impact of fluorinated liquids, the large-scale application of immersion cooling technology faces significant limitations. Consequently, oil coolants, which are more affordable and less volatile, have attracted considerable attention. Common oil coolants used in immersion cooling include mineral oils, synthetic oils, silicone oils and vegetable oils. Silicone oil has previously been demonstrated to offer certain advantages as an immersion coolant in photovoltaic cells [69,70], and in recent years, it has also been studied for immersion cooling in data centers [52,61,71]. Unlike the hydrocarbon structures of mineral and synthetic oils, silicone oil belongs to the class of organosilicon polymers. However, due to its oil-like properties, it is categorized as an oil coolant in discussions.

In terms of heat transfer performance, oil coolants exhibit advantages such as high specific heat capacity and high thermal conductivity. Shinde et al. [72] reported a PUE range of 1.03–1.17 for an immersion cooling system using mineral oil as the working fluid. Compared to air cooling, server power consumption was reduced by 16.83 %. Zhang et al. [73] compared the cooling performance of fluorinated liquid Noah3000A and mineral oil PAO-4 at the chip level. Due to its significantly higher thermal conductivity compared to conventional fluorinated liquids, mineral oil demonstrated slightly better cooling performance under low flow conditions. Wang et al. [61] found that the synthetic oil (XIAMETER™ 561) exhibited slightly better thermal performance than mineral oil (Therminol VP-1) due to its higher density and thermal conductivity. However, silicone oil, with its high viscosity, generally showed inferior cooling performance compared to mineral oil and synthetic oil in flow-based immersion cooling studies [52,61].

Oil coolants, particularly natural oils such as mineral oil, contain higher levels of impurities, leading to uncertainties regarding their impact on the reliability of IT equipment at various levels [74]. Compared to fluorinated liquids, oil-based coolants require more thorough compatibility evaluations, particularly with materials such as PCBs, PVC, electronic packaging, and passive components. Existing experiments indicate that oil-based coolants exhibit good compatibility with these key components in short- to medium-term immersion cooling applications. Pambudi et al. [54] reported that CPU components remained undamaged after being immersed in mineral oil for five months. Similarly, Shah et al. [75] conducted a six-month mineral oil immersion experiment, which revealed no incompatibility issues with PCBs or electronic packaging. However, they observed rapid aging of PVC casings, evidenced by a significant increase in Young's modulus, along with fading of printed labels on electronic components upon contact with oil (Fig. 5(a)). In a subsequent eight-month study, Shah et al. [76] reported adhesive and ink erosion in servers immersed in mineral oil (Fig. 5(b)(c)). For ester dielectric fluids, Van et al. [77] tested a novel biodegradable bio-based dielectric ester coolant (L795) under SPIC conditions. No significant compatibility issues were observed during an eight-month testing period.

Compared to mineral oils, synthetic oils undergo stricter manufacturing processes, containing minimal sulfur, aromatics, and nitrogen impurities. They also typically exhibit superior oxidation resistance and material compatibility. Bhandari et al. [78] and Bansode et al. [79] immersed halogen-free substrate cores in synthetic oils (EC-100, EC-110, and PAO-6) at temperatures ranging from 22 °C to 125 °C for 720 h, observing no significant compatibility issues. Shah et al. [80] conducted accelerated thermal aging tests following ASTM standards,

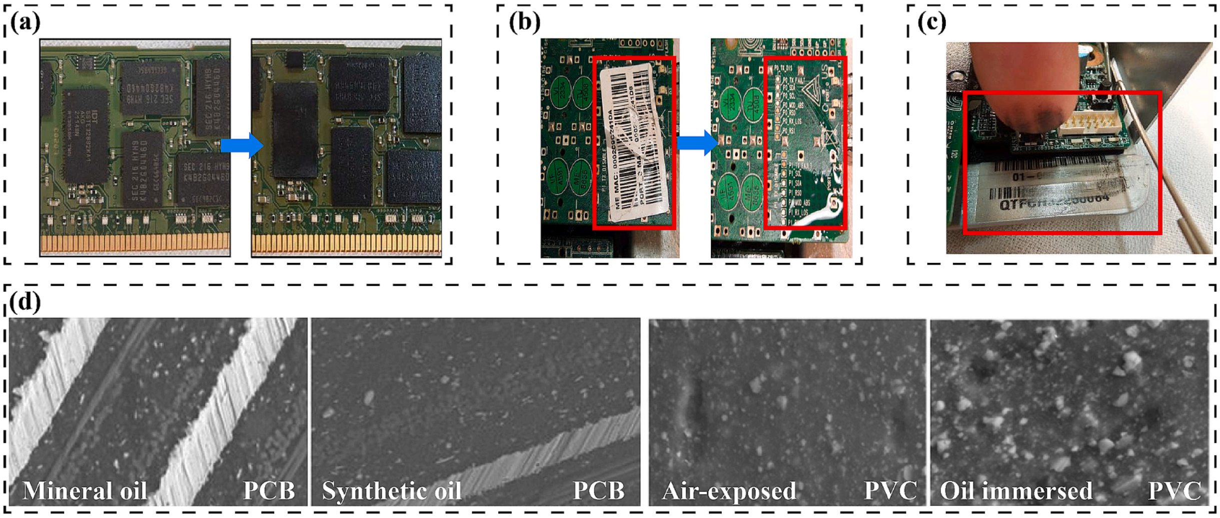


Fig. 5. Material compatibility testing of oil coolants: (a) Component labeling fading [75], (b) Adhesives concerns [76], (c) Ink/ Labeling issues[76], (d) Scanning images of PCB and PVC [80].

using mineral oil and EC-100 as immersion media. As shown in Fig. 5(d), scanning electron microscope analysis indicated that while PCB surfaces showed no cracks or penetration, PVC optical cable casings developed noticeable micropores and molecular aggregation, leading to hardening. However, no cracks or penetration were observed in passive components and synthetic rubber seals. Van et al. [77] evaluated a novel biodegradable bio-based dielectric ester coolant (L795) over an eight-month period, during which no compatibility issues were observed in PCBs, cables, or seals. The cables also exhibited no hardening, although adhesive erosion was again reported. Chauhan et al. [81] found that PCBs immersed in EC-100 at temperatures ranging from 22 °C to 105 °C for 720 h exhibited no significant change in thermal expansion coefficient. Ramdas et al. [82] studied the long-term effects of synthetic oil immersion by submerging different types of PCBs in EC-100 for one year. Their findings showed a reduction in Young's modulus by 33.33 % for 370HR PCBs and 19.04 % for 185HR PCBs. This reduction in modulus enhances PCB flexibility, potentially reducing solder joint failures and ultimately extending the lifespan of electronic packaging. Overall, while oil-based coolants exhibit lower material compatibility than fluorinated liquids, they have demonstrated satisfactory compatibility with key server components, particularly printed circuit boards (PCBs). However, the effects on PVC materials, adhesives, printed inks, and component markings require careful consideration. Compatibility with various plastics and rubber materials should be further evaluated. For additional material compatibility data, the Open Compute Project reports can be referenced [63].

The signal compatibility of oil-based coolants largely depends on their dielectric constant. Lo et al. [83] compared the high-speed signal integrity of servers designed for air cooling in air, FC-40 (Dk = 1.91), and synthetic oil POE 390z (Dk = 3.4). The study found that microstrip signals are significantly affected by immersion cooling fluids. The lower dielectric constant of FC-40 has a milder impact on signals, whereas at a frequency of 16 GHz, the insertion loss of POE 390z, with a higher dielectric constant, is 0.03 to 0.44 dB per inch higher than in an air-cooled environment, and the impedance decreases by 4 to 8 Ω, though still within the standard requirements. In the studies on signal compatibility of oil and fluorinated coolants, although the majority of the basic parameters for signal integrity meet the specified limits, higher dielectric constants result in greater signal loss. Therefore, in SPIC cabinets equipped with high-frequency devices, coolants with lower dielectric constants should be prioritized. Additionally, optimization of critical components such as PCB impedance, connectors, and cables is necessary.

In terms of environmental friendliness, vegetable oils [84] and ester-based synthetic oils [85] generally exhibit good biodegradability, while only some hydrocarbon-based synthetic oils are biodegradable. In contrast, the biodegradability of typical mineral oils does not exceed 30 % [86]. In terms of cost, synthetic oil is priced higher than mineral oil but lower than fluorocarbon fluids. The price of synthetic oil coolant is approximately twice that of mineral oil, while the unit price of fluorinated fluids can reach 5 to 10 times the unit price of synthetic oil [87].

2.3. Comparison of various single phase immersion coolants

High-performance coolants must exhibit excellent thermophysical properties, material compatibility, chemical and thermal stability, non-corrosiveness, and environmental friendliness. The selection of high-performance SPIC coolants is ideally guided by the following criteria: specific heat capacity >960 J/(kg·K), thermal conductivity >0.06 W/(m·K), dielectric constant < 2.3, boiling point >100 °C, flash point for oil coolants >150 °C, and GWP < 250 [63,88,89].

To optimize the selection of cooling liquids, the concept of a Figure of Merit (FOM) was introduced to enable the quantitative comparison of their thermophysical properties. An ordinary FOM is typically derived from the heat transfer coefficient correlation by neglecting terms unrelated to the fluid properties. The ordinary FOM expression applied to SPIC in current studies is as follows [62,90]:

$$FOM = \frac{\lambda^a c_p^b \rho^c}{\mu^d}. \quad (1)$$

here, λ represents the thermal conductivity of the fluid, C_p denotes the specific heat capacity, ρ is the density, and μ is the viscosity. The coefficients a, b, c and d vary depending on the turbulence model employed. A higher FOM generally indicates superior thermal performance of the coolant. In practical engineering applications, the coolant in pump driven SPIC cabinets typically exhibits turbulent flow. If the classic Dittus-Boelter equation is used, the FOM can be expressed as follows:

$$FOM = \frac{\lambda^{0.6} c_p^{0.4} \rho^{0.8}}{\mu^{0.4}}. \quad (2)$$

Key parameters for selecting coolants include viscosity, density, thermal conductivity, dielectric constant, specific heat capacity, boiling point, and flash point [91]. Table 1 summarizes these critical properties for commonly used oil coolants and fluorinated fluids in SPIC literature,

Table 1
Single-phase immersion cooling medium used in some literature.

Type	Name	25℃ Densitykg/m3	35℃ ViscositymPa·s	Specific heat capacityJ/(kg·℃)	Thermal conductivityW/(m·℃)	FOM×10^3	Boiling point℃	Flash point ℃	Dielectric constant	GWP	The references used
Bio-based dielectric ester coolant L759		898(20℃)	6.38(40℃)	1960	0.145	0.72		192	3.51		[77]
Silicone oil	XIAMETER™ 561	957	42	1491	0.143	0.31		> 300	2.7	Low	[61,92]
Mineral oil		849	11	2198	0.13	0.54		≈185	≈2.1	Low	[93,94]
Synthetic oil	PAO-6	817	29	2377	0.145	0.39		239		Low	[48]
	PAO-4	798	17	2143	0.15	0.47		222	2.2	Low	[73]
	PAO-2	796	4.1(40℃)	2300	0.15	0.84		159			[95]
	EC-100	804	10.6(30℃)	2165	0.138	0.54					[96]
	EC-110	820(20℃)	5.95(40℃)	2212	0.137	0.69		203	2.1	Low	[97]
	SmartCoolant	796(20℃)	4.06(40℃)	2260	0.14	0.81					[98,99]
	OptiCool 872552	798(16℃)	4(40℃)	2204	0.136	0.79		185			[100]
	Therminol VP-1	1060	3.38	1597	0.135	0.93		124	3.35		[61,101]
Fluorinated liquid	FC-40	1840	3	1061	0.065	0.83	165	-	1.9	9000	[48]
	FC-43	1860	2.8	1100	0.065	0.87	174	-	1.9	> 5000	[52]
	FC-3283	1816(35℃)	1.4	1047	0.064	1.10	128	-	1.9	> 5000	[93]
	Noah®3000A	1787	2	1159	0.0614	0.95	110	-	1.79	120	[73]
	HFE 7300	1660	1.18(25℃)	1140	0.069	1.18	98			200	[102]
	HFE 7500	1614	1.1	1125	0.065	1.14	128	-	5.8	100	[103]
	HFE 7700	1797	3.1	1040	0.064	0.79	167	–	6.7	420	[104]

includes FOM calculated from Eq. (2). The table reveals that oil coolants exhibit significantly higher specific heat capacity and thermal conductivity, while fluorinated fluids have lower viscosity and higher density. It is also observed that oil coolants and PEC coolants generally have lower dielectric constants, whereas HFE fluids show notably higher dielectric constants, which could potentially affect signal transmission in electronic devices. The FOM of water at 35 °C is approximately 5–8 times higher than that of fluorinated liquids, 6–12 times higher than that of mineral oils and synthetic oils, while silicone oils generally exhibit an FOM that is only half that of mineral oils. This provides a preliminary analysis of the differences in thermal performance among various coolants. It is worth noting that this type of FOM is related to the turbulent Nu number expression, meaning that under different flow conditions, the cooling performance of fluorinated liquids and oil-based coolants may vary. However, overall, using a quantitative approach to compare the properties of coolants aids in selecting the most suitable coolant for immersion cooling systems.

To improve the thermophysical properties of immersion cooling fluids, some researchers have investigated the cooling performance of nanofluids in SPIC. Nanofluids are colloidal suspensions in which nanoparticles are uniformly dispersed in a base fluid [105]. Studies have shown that the high thermal conductivity of nanoparticles, their enhancing effect on microscale convection, and the interactions between particles can significantly improve the thermal conductivity and convective heat transfer coefficient of immersion cooling fluids [106,107]. However, current nanofluids face stability issues, as their physical properties remain constant only for short periods. Over time, nanoparticle aggregation can occur, leading to reduced thermal conductivity and increased viscosity [108], eventually forming agglomerates that settle. After prolonged use, this settling may accumulate at the bottom of the cabinet, resulting in a loss of heat transfer enhancement and even causing blockages in the piping, severely affecting system reliability. Furthermore, the material compatibility of nanofluids with internal server components remains unclear, and conductive nanoparticles can compromise the insulation properties of the cooling fluid [109]. Existing research on nanofluid-enhanced SPIC heat dissipation primarily focuses on chip-scale heat dissipation performance and pressure losses, making it difficult to assess the risks associated with stability and compatibility. Therefore, the practical application of nanofluids in data center SPIC systems requires a focus on overcoming stability challenges and conducting cabinet-level experiments to comprehensively evaluate their long-term impact on electronic devices.

3. Chip level heat transfer enhancement

Fig. 6 illustrates the trends in transistor count, maximum power consumption, and heat flux of electronic chips over the past two decades. It is evident that the power density of electronic devices has grown almost exponentially. To address the thermal challenges posed by the increasing heat flux density of electronic devices, SPIC systems must adopt measures to enhance the cooling performance for critical high-heat-flux components. For high-heat-flux chips, it is often necessary to implement various active and passive heat dissipation enhancement techniques to meet the thermal management requirements. This section reviews the passive heat dissipation enhancement methods related to the design and configuration of SPIC heat exchangers, as well as active

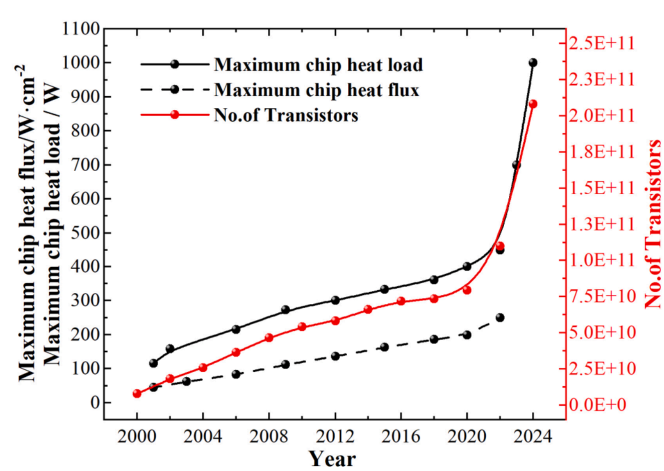


Fig. 6. Variation in the number of transistors, maximum heat flow and heat flux of an electronic chip [110].

cooling strategies aimed at addressing high heat flux scenarios.

3.1. Heat sink structure optimization

Using heat sinks is an effective way to expand the heat dissipation area and enhance heat transfer performance. Given the significant differences in viscosity and thermophysical properties between immersion coolants and air, heat sinks should be specifically designed for SPIC applications. For instance, Sarangi et al. [111] designed a copper finned heat sink with thicker fins (0.8 mm) compared to traditional air-cooled heat sinks to improve fin efficiency. Similarly, Wu et al. [112] demonstrated that under forced convection SPIC, a louvered fin heat sink with thicker 0.5 mm fins exhibited superior thermal performance compared to 0.3 mm fins.

Numerous factors influence the effectiveness of immersion liquid cooling. Multi-objective optimization can be employed to comprehensively evaluate these factors. For the straight finned heat sink shown in Fig. 7(a), Li et al. [113] used an orthogonal analysis method under forced convection conditions and ranked the importance of parameters affecting the maximum temperature as follows: fin thickness, fin count, fin height, and baseplate thickness. Li et al. [114] investigated the effects of seven parameters on forced convection SPIC using the response surface methodology. Their study revealed that the interaction between fin spacing and flow rate, as well as fin height and flow rate, significantly impacts the maximum temperature of the heatsink. Suresh et al. [115] optimized the thermal management of a multi-chip system under forced convection SPIC by adjusting the amplitude and wavelength of a wavy fin heat sink. Compared to straight fins, a wavy fin heat sink (As shown in Fig. 7(b)) with an amplitude of 0.4 mm and a wavelength of 1 mm reduced the average and maximum temperatures by 3.9 °C and 5.3 °C, respectively. The research results indicate that standard SPIC heat sink designs should incorporate slightly thicker fins than air-cooled heat sinks to accommodate the higher viscosity of immersion coolants, and improve fin efficiency. Given the complexity of immersion environments, the strategic use of numerical simulations and multi-objective optimization methods can effectively enhance the thermal performance of heat sinks.

Compared to conventional straight-fin heat sinks, pin–fin heat sinks are generally considered to have superior turbulence-inducing capabilities and provide a larger heat dissipation surface area within the same volume, thereby enhancing heat transfer efficiency [116]. The adoption of pin–fin structures results in lower thermal resistance, with heat transfer enhancement reaching 1.6 to 2 times that of finned surfaces under certain conditions [117]. Soni [118] investigated the optimal parameters for plate-fin and pin–fin heat sinks under forced convection oil immersion conditions at a flow rate of 1 L/min. His study suggested immersion heat sink dimensions with thicknesses or diameters ranging from 0.4 to 0.6 cm and heights from 2.5 to 3.5 cm. One of the pin–fin heat sinks is shown in Fig. 8(a). In immersion liquid cooling, pin–fin heat sinks also effectively reduce pressure drop. Niazmand et al. [107] investigated the effect of Al2O3 oil-based nanofluids with concentrations of 0 % and 5 % by mass on the performance of three types of heat exchangers: finned, pin–fin, and plate-fin, at flow rates ranging from 0.5 to 3 LPM. The study results showed that the mass concentration of the nanofluid had little impact on pressure drop but could significantly enhance heat dissipation performance. For the pin–fin heat exchanger, under conditions of 35 °C, 3 LPM, and a 5 % mass concentration, a 55 % reduction in pressure drop was achieved at the cost of only an 11 % higher temperature compared to the finned heat exchanger. However, solid pin fins may cause flow separation from the surface, forming stagnant flow zones behind the fins, which increases thermal resistance [119]. To mitigate this issue, perforated and slotted pin–fin structures can be employed. As depicted in Fig. 8(b), Zhu et al. [120] designed two types of slotted pin–fin structures for SPIC: one with a central slot and another with diagonal slots. Compared to regular pin fins, the square pin–fin heat sinks with central and diagonal slots increased the Nusselt number by 13.98 % and 19.69 %. Under SPIC conditions, pin–fin heat sinks exhibit significant potential for both heat dissipation and energy efficiency. However, their higher manufacturing cost is a notable drawback.

In addition to conventional finned and pin–fin heat sink designs, recent studies have explored advanced manufacturing techniques to develop heat sinks with novel and complex structures. As shown in Fig. 8 (c), Lamotte et al. [97] employed an electrochemical additive manufacturing (ECAM) method, which offers a relatively low-cost approach, to fabricate lattice-structured heat sinks with various wall thicknesses and porosities. Among these, a heat sink with dimensions of 90 × 70 × 21 mm (L × W × H), a porosity of 68 %, and a wall thickness of 1.2 mm effectively cooled a 250 W chip under natural convection SPIC conditions. Similarly, Herring et al. [121] utilized the ECAM technique to manufacture a body-centered cubic (BCC) lattice heat sink with dimensions of 25 × 25 mm, a lattice height of 21 mm, and a lattice thickness of 1.2 mm. Under forced convection, this heat sink achieved a 12 %–34 % temperature reduction compared to a conventional finned heat sink with the same volumetric occupancy within a porosity range of 60 %–80 %. As manufacturing technologies continue to advance, the production of heat sinks with complex structures will become more feasible. This type of heat sinks typically exhibit a higher surface-area-to-volume ratio, which enhances the maximum heat dissipation capability in confined immersion environments.

3.2. Heat pipes and vapor chamber heat sinks

When liquid-cooled metal-based heat sinks are used for high heat flux electronic devices, the thermal conductivity of the heat sink material itself may limit cooling performance. This is particularly true when heat transfer enhancement relies on expanding the heat dissipation area, as using pure metal as the conductive substrate may lead to severe

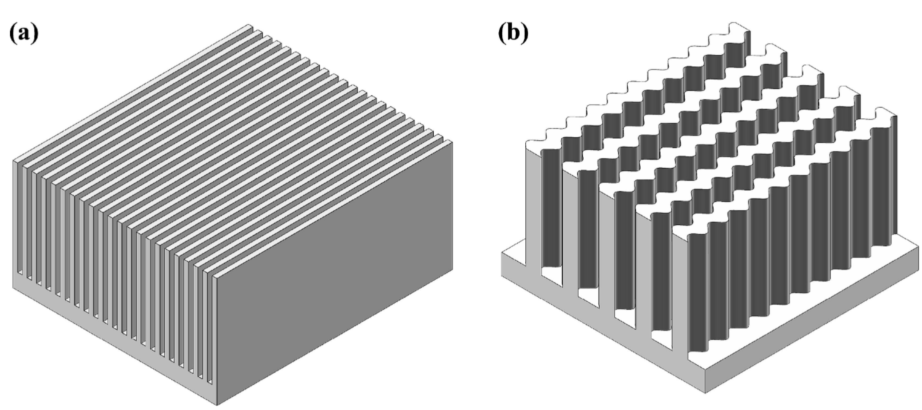


Fig. 7. SPIC finned heat sink: (a) straight finned heat sink [113], (b) wavy fin heat sink [115].

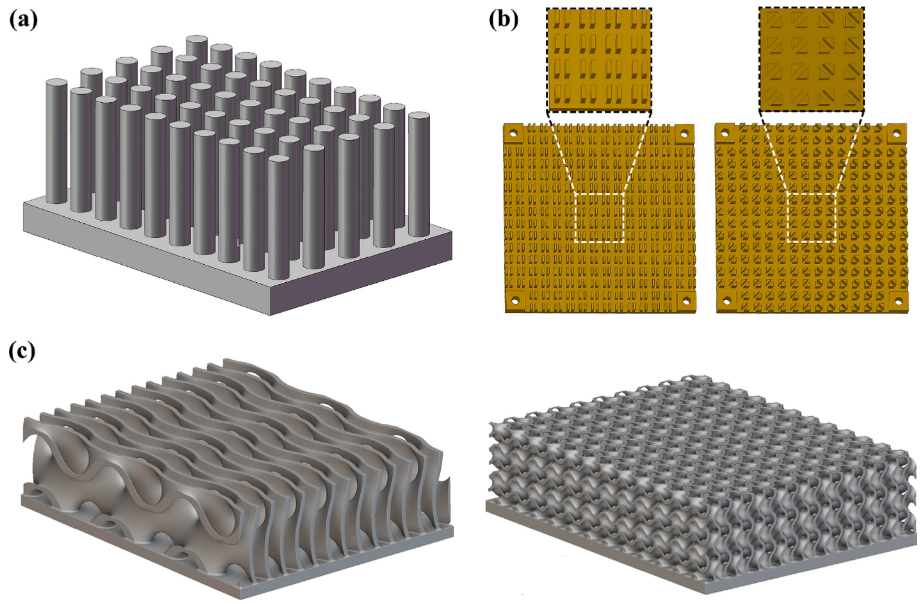


Fig. 8. The schematic of SPIC heat sinks: (a) pin fin [118], (b) slotted pin–fin structures [120], (c) lattice structures [97].

temperature uniformity issues, hindering effective area expansion. In this case, using heat pipes and vapor chambers (VCs) with extremely high equivalent thermal conductivity is an excellent choice.

Currently, researchers have explored the use of heat pipes and vapor chambers in SPIC systems [122,123]. Muneeshwaran et al. [124] compared the thermal performance of three copper finned heat sinks with identical fin dimensions and a base size of 118 × 78 mm: a conventional metal base, a base embedded with heat pipes, and a VC-based heat sink. The heat pipe-embedded base and VC-base heat sinks are shown in Fig. 9(a)(b). Compared to the metal base heat sink, the heat sinks with embedded heat pipes and VC bases reduced thermal resistance by approximately 13 % and 19 %, achieving values of 0.072 K/W and 0.067 K/W, respectively. As shown in Fig. 9(c), Luo et al. [125] applied a four-heat-pipe finned heat sink in a forced convection SPIC system. The results indicated that, compared to a conventional square-finned heat sink, the heat pipe-based heat sink—offering a larger effective heat exchange area—achieved higher heat transfer efficiency, improved temperature uniformity, and lower node temperatures. Luiten [126] designed and optimized a VC-based heat sink measuring 200 × 200 mm with 80 fins of 20 mm height. Under natural convection cooling, this heat sink maintained a 500 W chip at approximately 85 °C, achieving a minimum total thermal resistance (R_{jf}) of only 0.089 K/W. Compared to a 100 × 100 mm VC heat sink under the same conditions, the total thermal resistance was reduced by approximately 0.06 K/W. By leveraging the high thermal conductivity of heat pipes and VCs, these heat sinks significantly increase the effective heat dissipation area in high expansion ratio designs compared to pure metal heat sinks, demonstrating great potential for cooling high heat flux density chips.

3.3. Active convection heat transfer enhancement

Due to the limited internal space of servers, passive cooling methods alone are difficult to meet the thermal demands of future kW-level chips. Additionally, in conventional SPIC systems, cooling liquid typically enters from the side of the server, causing a significant portion of the fluid to bypass the heat sink fins and flow through bypass channels (as

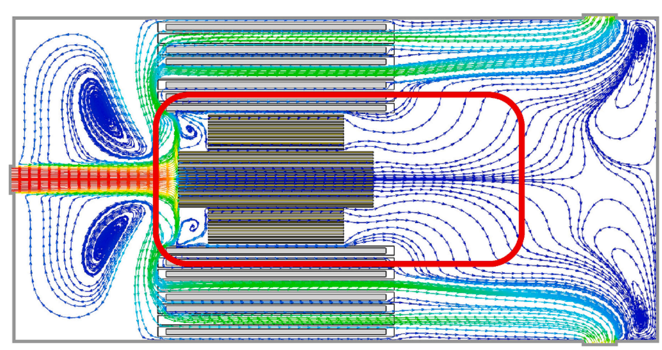


Fig. 10. Schematic of the effect of bypass on coolant flow [73].

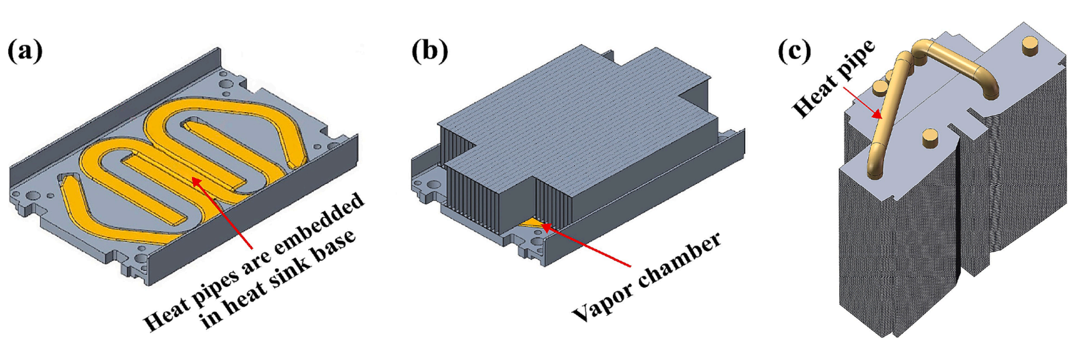


Fig. 9. Heat pipes and vapor chamber heat sinks: (a) heat pipe-embedded base heat sink [124], (b) vapor chamber base heat sink [124], (c) heat pipe finned heat sink [125].

illustrated in Fig. 10), thereby reducing the heat dissipation effectiveness of the heat sink [127]. Sarangi et al. [128] found that under forced convection cooling with a total flow rate of 7.3 L/min in a standard 1U chassis, only about 8 % of the flow passed through the CPU heat sink. This bypass effect becomes particularly detrimental to heat transfer performance when the coolant has high viscosity or when the inlet flow rate is large [48]. Therefore, active cooling enhancement techniques are required to increase the coolant flow through the heat sink, thereby improving overall heat dissipation efficiency.

Currently, there are two primary approaches for active convection enhancement in SPIC. The first approach involves direct-to-chip cooling, similar to cold plate liquid cooling. In this method, the coolant entering the server first cools high heat flux chips before dispersing to cool other heat-generating components within the server. Alternatively, the process can be reversed, where the coolant exits through the chip heat sink after circulating within the system. Zhang et al. [73] designed an open cold plate heat sink, which features fin structures similar to conventional liquid-cooled cold plates but with additional openings on both sides, as shown in Fig. 11(a). After entering the cold plate, the immersion coolant first cools the chip before flowing into the immersion environment through these openings. Their results indicated that, using mineral oil as the working fluid, the open cold plate reduced thermal resistance by approximately 50 % compared to a conventional finned heat sink with over ten times the heat dissipation surface area. Khalili et al. [129] introduced EC-110 dielectric liquid at 30 °C into a 25 × 25 mm cold plate with fin dimensions of 0.1–0.2–2 mm (t-p-H). When the coolant experienced a temperature rise of 5 °C between inlet and outlet, the system was capable of dissipating 987 W of heat from a chip of the same size. Simon et al. [130] similarly optimized a single-phase submerged liquid-cooled cold plate by directly passing EC-100 dielectric fluid inside a normal liquid-cooled cold plate, with the lowest thermal resistance (R_{cf}) of only 0.055 K/W. Saini et al. [131] designed an embedded manifold heat sink by laterally inserting manifolds into a conventional heat sink, directing coolant through the fins for enhanced heat dissipation, as shown in Fig. 11(b). The heat sink achieved a minimum thermal resistance of approximately 0.045 K/W and maintained pressure losses below 1 kPa at flow rates under 4 L/min. Intel's G-flow scheme established a connected pipeline between the chip heat sink and the cabinet liquid outlet, using gravity or pump suction to accelerate coolant flow through the heat sink [132].

The second type is the internally driven flow mode, which actively modifies the internal flow field of the server to ensure that the majority of the coolant flows through the chip heat sink. Some designs incorporate localized jet impingement to enhance turbulence. Intel and Submer introduced a forced convection heat sink, which integrates a shell channel around a conventional finned heat sink. A propeller at the front of the channel draws immersion coolant into the heat sink, increasing flow velocity through the fins, which achieving reported cooling capabilities of up to 1000 W [133]. As illustrated in Fig. 11(c). Cai et al. [134] proposed a porous nozzle cooling scheme. With a flow rate of only 0.55 L/min, the scheme achieved an average heat transfer coefficient of 1182.61 W/(m^2·K) using mineral oil and 5368 W/(m^2·K) using fluorinated liquid as the working fluid on a chip without a heatsink. Recently, Azarifar et al. [135] developed a synthetic jet device for SPIC utilizing a piezoelectric diaphragm, as shown in Fig. 11(d)). Driven by a DC voltage, the diaphragm oscillates, inducing high-frequency volume changes within the cavity to generate fluid jets. This device demonstrated significant heat transfer enhancement with only milliwatt-level power consumption. Furthermore, research on using ultrasonic methods to enhance SPIC cooling performance is also underway [136].

Table 2 summarizes studies on single-phase immersion chip-level cooling with high heat flux. Among various heat dissipation enhancement methods, metal-based heat sinks are suitable for scenarios with limited space and moderate heat flux. Heat pipe and VC heat sinks require a large area expansion ratio to achieve significant heat transfer enhancement, making them highly applicable in environments with large spaces and high heat flux. Forced convection heat sinks can dissipate heat from high-power electronic devices within a compact volume but generally involve more complex piping and heat sink structures. Notably, the design of immersion cooling heat sinks should prioritize fin efficiency to ensure effective heat dissipation. Furthermore, different enhancement methods can be combined to maximize cooling performance. It is foreseeable that integrating active convection with a VC can achieve cooling performance exceeding 1 kW & 100 W/cm^2, meeting the long-term cooling demands of high-end chips in the future. For these high-heat-flux chips, existing SPIC studies have shown that non-uniform power distribution significantly impacts heat dissipation performance [129]. The heat generated by real chips is non-uniform

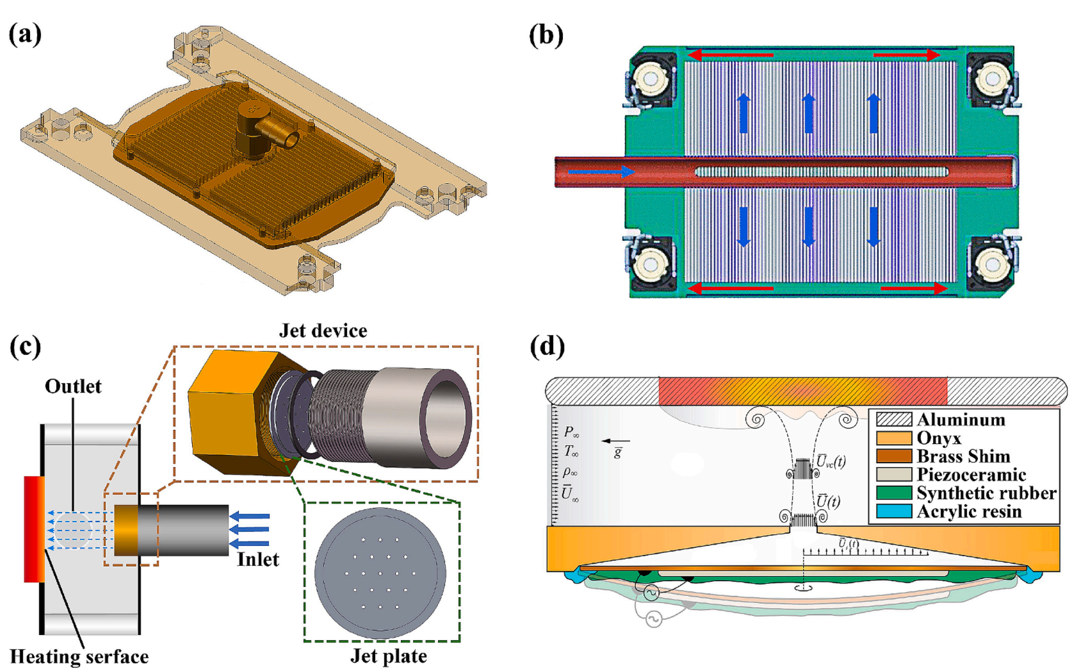
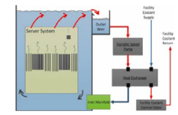
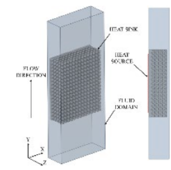
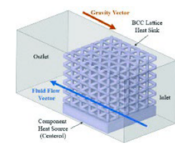
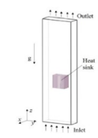
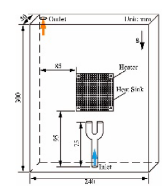
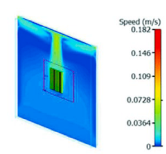
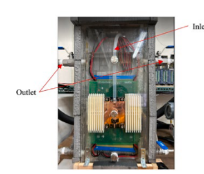
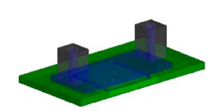
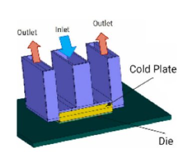
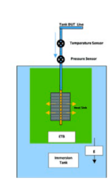


Fig. 11. Active forced convection heat sink: (a) embedded manifold [73], (b) open cold plate [131], (c) multi-jet [134], (d) synthetic jet device [135].

Table 2
Summary of research on Single-phase immersion high power chip cooling.

Reference	Technical description	Working fluid	Condition	Thermal resistance, K/W Base size: L × W, mm Fin size: t-p-H, mm	Schema
Sarangi[111]	Natural convection cooling Copper Finned heat sink	PAO-6	$T_f = 40$ °C $Q_{max} = 270$ W	$R_{cf} = 0.135$ 24 fins, 0.8 mm thickness, 21 mm high	
Lamotte[97]	Natural convection cooling Copper Lattice structures heat sink	EC-110	$T_f = 40$℃ $Q_{max} = 250$ W $q_{max} = 5.7$ W/cm^2	$R_{bf} = 0.126$ 90 × 70 Wall thickness –high: 1.2–21 Porosity: 68 %	
Herring等[121]	Forced convection cooling Copper Body-centered cubic lattice heat sink	PAO-6	$T_{in} = 40$℃ $Q_{max} = 150$ W $q_{max} = 66.7$ W/cm^2	$R_{jf} = 0.2$ 25 × 25 Wall thickness – high: 1.2–21	
Liet al.[113]	Forced convection cooling Aluminum Finned heat sink	FC-40	$T_f = 27$ °C $Q_{max} = 300$ W $q_{max} = 9.9$ W/cm^2	$R_{bf} = 0.103$ 75 × 65 1.5–1.7–30	
Zhu et al.[120]	Forced convection cooling Copper Slotted pin–fin structures	FC-40	$T_f = 26.85$ °C Re = 225 ~ 720 $Q_{max} = 180$ W	$R_{bf} = 0.15$ 70 × 70 2.5–2.5–4	
Luiten[126]	Natural convection cooling Copper base, aluminum fins vapor chamber finned heat sink	Oil	$T_f = 35$ °C $Q_{max} = 500$ W $q_{max} = 22$ W/cm^2	$R_{jf} = 0.089$ $R_{cf} \approx 0.04$ 200 × 200 80 fins, 20 mm high	
Zhang et al.[73]	Active convection heat transfer enhancement Copper Open cold plate heat sink	PAO-4 Noah®3000A	$V_f = 1 \sim 3$ LPM $Q_{max} = 300$ W	PAO-4: $R_{cf} = 0.05$ Noah®3000A: $R_{cf} = 0.041$ 60 × 43.7 0.2–0.6–3	
Simon et al.[130]	Active convection heat transfer enhancement –- Cold plate heat sink	EC-100	$T_f = 40$ °C $V_f = 3$ LPM $Q_{max} = 400$ W $q_{max} = 16$ W/cm^2	$R_{bf} = 0.025$ $R_{cf} = 0.055$ 55 × 55 1.4–1-5–4.5	
Khalili等人[129]	Active convection heat transfer enhancement copper Cold plate heat sink	EC-110	T_{in}-$T_{out} = 5 \sim 20$℃ $Q_{max} = 987$ W $q_{max} = 152$ W/cm^2	$R_{jf} \approx 0.058$ 25 × 25 0.2–0.1–2	
Saini et al.[131]	Active convection heat transfer enhancement Copper base, aluminum fins embedded manifold heat sink	PAO-6	$T_f = 40$℃ $V_f = 0.8 \sim 8$ LPM $Q_{max} = 400$ W $q_{max} = 12.4$ W/cm^2	$R_{cf} = 0.045$ 80 fins, 1 mm pitch	

and unstable, yet current research often treats chips as uniform heat sources. To better reflect the actual thermal management conditions of chips, using appropriate thermal simulation methods to model the dynamic thermal distribution of chips is an effective approach [137].

4. Server and cabinet level optimization strategies

In addition to the high heat flux chips that generate the majority of the heat, other heat-producing components such as memory, hard drives, and power supplies are present within data center servers. Designing single-phase immersion-cooled servers requires balancing the cooling demands of these chips with other heat-generating components while minimizing pressure drop. Furthermore, to prevent improper flow distribution among servers in the same liquid-cooled cabinet, the cabinet must feature appropriate flow distribution structures to ensure overall cooling performance. This section reviews flow channel optimization structures within servers, flow distribution designs for liquid-cooled cabinets, and several heat transfer enhancement strategies implemented inside servers.

4.1. Server structure optimization

The flow path of immersion cooling fluid within a server significantly impacts the thermal performance of heat-generating components and the associated flow resistance of the system. Adjustments to the inlet and outlet configurations, addition of baffles, and incorporation of flow distribution devices can effectively modify the fluid flow pathways and reducing bypass flows, thus enhancing cooling efficiency. Eiland et al. [138] demonstrated that directing the coolant to flow directly over chip heat sinks reduced chip temperatures by 5 °C. Shrigondekar et al. [48] investigated various inlet–outlet configurations for single-server cooling, as shown in Fig. 12(a), including T-shaped, U-shaped, and Z-shaped arrangements. The study revealed that the T-shaped configuration outperformed the Z-shaped and U-shaped designs, achieving lower thermal resistance and reduced temperatures. Baffles within the server can further optimize flow paths and reduce bypass flow. As shown in Fig. 12 (b), Huang et al. [139] swapped the inlet and outlet positions located at the top and bottom of the server, examining the effects of pro-gravity and anti-gravity flow directions on heat transfer performance and pressure loss. Results showed that under anti-gravity flow, the chip casing temperature decreased by 33.8 %, while pro-gravity flow improved cabinet temperature uniformity by 32.5 % and reduced system pressure drop by 9.8 % at the same flow rate. Li et al. [114] introduced baffles on both sides of server chips and RAM, with the baffle placement shown in Fig. 12(c). The study found that increasing the baffle height reduced chip temperatures, especially at low flow velocities. As exemplified in Fig. 12(d). Qi et al. [140] designed a flow distributor with elongated slots and square holes to ensure uniform fluid distribution. However, despite achieving uniform flow, the study revealed that hotspot temperatures within the server remained 24–40 °C higher than those of other electronic components.

To address local hotspots in high-heat-flux electronic components within servers, Ding et al. [141] proposed two techniques: turbulence enhancement using micro-pumps (Fig. 13a) and flow redistribution

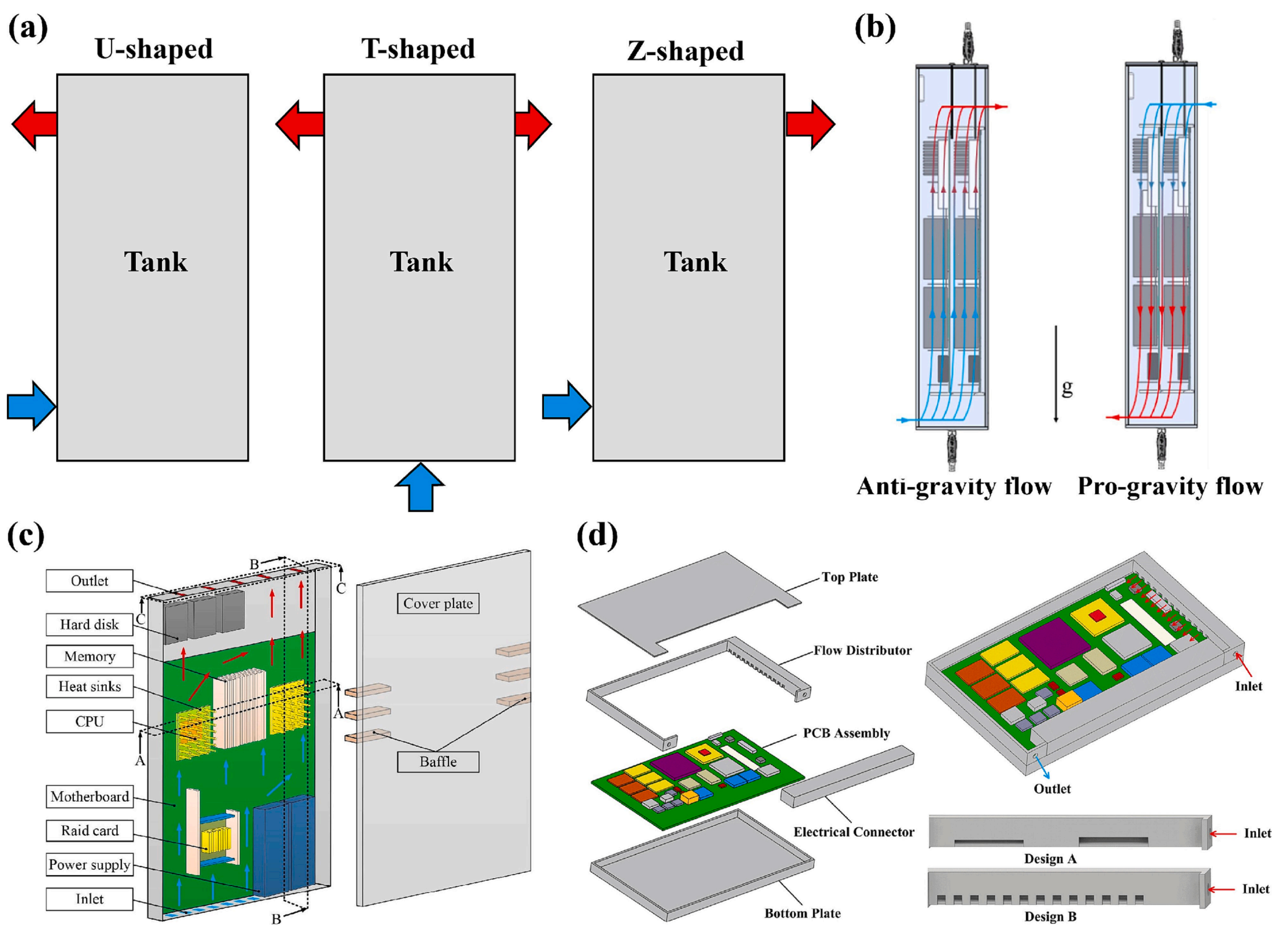


Fig. 12. The Strategies of Immersion Cooled Server Flow Path Optimization: (a) import/export arrangement [48], (b)anti/pro gravity flow[139], (c) addition of baffles[114], (d) flow distributor[140].

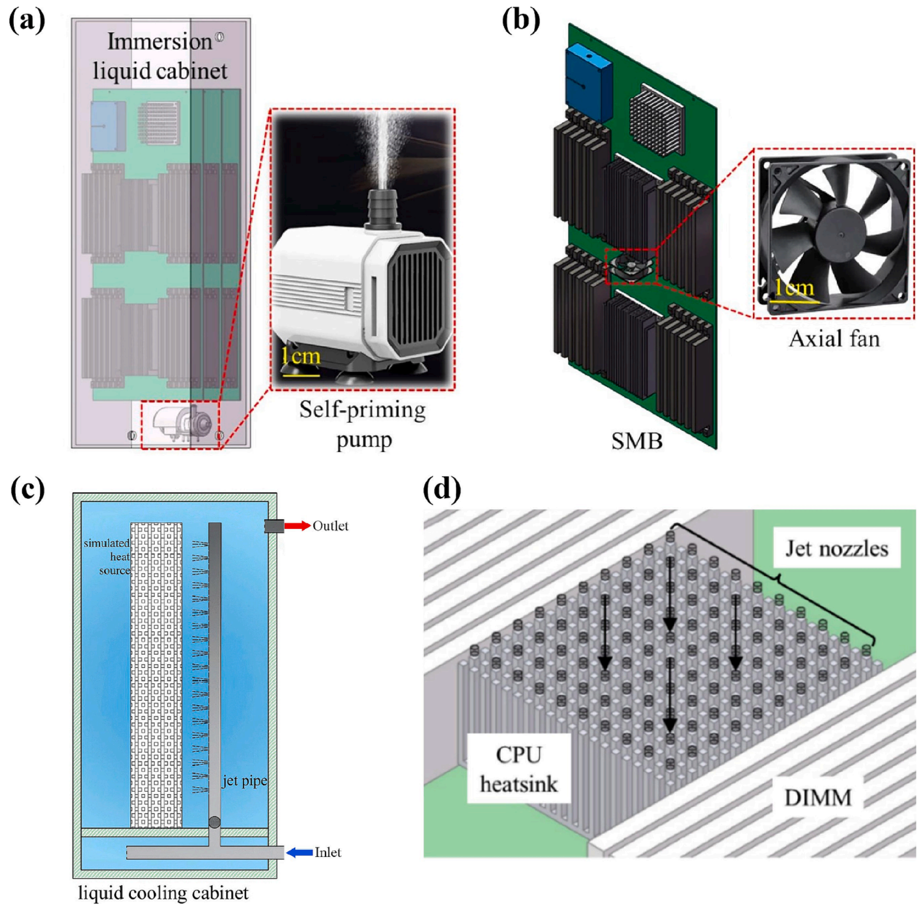


Fig. 13. Server-level heat dissipation enhancement strategies: (a) micro-pumps [141], (b) flow redistribution using fans [141], (c) large-area jet [142], (d) local jet assist [144].

using fans (Fig. 13b). These methods reduced the maximum CPU surface temperature by 6.1 % and 5.0 %, respectively. The micro-pump jet method enhanced flow turbulence within the server, significantly improving cooling efficiency and temperature uniformity for chips closest to the jet direction. While fans could enhance chip cooling, their suction effect on the coolant might deteriorate heat transfer at other locations. Yuan et al. [142] developed a large-area jet impingement cooling experimental system to enhance overall thermal performance in immersion servers. As shown in Fig. 13(c), this large-area jet impingement device is positioned outside the server, where jet nozzles are used to cool the entire server. Results showed that the surface heat transfer coefficient of the large-area jet impingement cooling system was 2.6 times that of a conventional immersion cooling system. Zhang et al. [143] employed a similar jet structure combining flow immersion and jet impingement cooling methods for server cooling. Using orthogonal analysis, they determined the optimal jet parameters and analyzed the cooling performance and pressure drop under Z-shaped and U-shaped flow immersion inlet conditions. As shown in Fig. 13(d), Liu et al. [144] proposed a dual-inlet jet-assisted SPIC system, which, in addition to the conventional coolant inlet, included a jet inlet located above the chip heat sink to enhance hotspot cooling. Compared with conventional SPIC servers, this approach increased the local *Nu* at the CPU by 85.5 %, under the condition of a coolant inlet temperature of 30 °C, the peak temperature was reduced from 55.5 °C to approximately 52 °C, significantly improved server temperature uniformity. The principles of turbulence enhancement and flow redistribution heat dissipation enhancement technologies are the same as those of the second type of active convective heat transfer enhancement techniques discussed in Section 2.3, and both can effectively improve the heat dissipation performance of hotspots. The subtle difference lies in that active heat dissipation enhancement methods at the server level have a greater impact on the overall flow field of the server, affecting the heat dissipation performance over a larger area.

To further reduce energy consumption and enhance localized cooling performance, Hnayno et al. [145] combined direct-to-chip water-cooled (WC) cold plate systems with passive SPIC cooling technology, proposing a hybrid SPIC/WC system with cold plates in series with cooling coils. as exemplified in Fig. 14. The study evaluated cooling performance and energy efficiency under various immersion media and water loop configurations. Results indicated that compared to liquid cooling systems, the SPIC/WC system reduced power consumption by at least 20.7 %. The study by Simon et al. [146] demonstrates that the temperature distribution of memory and chips is more uniform and stable in a hybrid cooling environment where WC cold plates and immersion cooling methods are employed simultaneously. This method, which combines WC cold plate with SPIC, not only effectively eliminates localized hotspots in servers through WC method, but also addresses the cooling needs of other low-heat-flux electronic devices through immersion cooling. Given the increasing chip heat flux density, this approach has certain reference value. However, it also increases system complexity. The use of liquid-cooled cold plates introduces the risk of leakage, which is more difficult to detect and maintain in an immersion environment. Therefore, the long-term economic viability and reliability of this solution still require further research.

4.2. Cabinet structure optimization

The performance of SPIC cabinets is often determined by the method

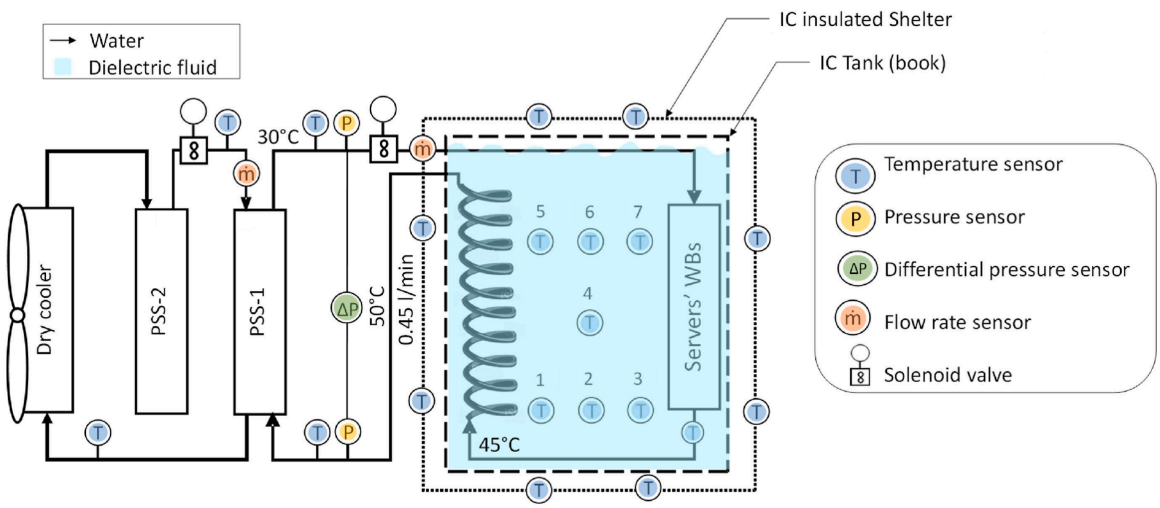


Fig. 14. Single-phase immersion/water-cooled cold plate technology [145].

of flow distribution employed. Effective flow distribution ensures that all servers within the cabinet receive adequate cooling, preventing scenarios where insufficient flow creates stagnation zones and leads to overheating in certain servers. Part of the cabinet without shunt chamber through the baffle angle adjustment and entrance and exit configuration to achieve the flow distribution Liu et al. [147] conducted optimization studies on such cabinet designs. The results demonstrated a 46.41 % improvement in overall temperature uniformity after optimization. Among the various factors analyzed, baffle angle was identified as the most significant influence on temperature distribution, followed by the number of baffles and the arrangement of inlets and outlets.

Forced convection SPIC systems with parallel server flow channels are often affected by interaction effects, where a change in flow rate through one channel can impact the flow rates in other branches. Utilizing a combination of pressure chambers and porous panel has been shown to effectively mitigate this issue. Luiten [148] conducted simulation studies on cabinets without flow distribution structures and found that removing one server chassis reduced the flow rate of the remaining server chassis by 90 %, leading to a 30 % decrease in cooling performance. However, adding a porous panel at the bottom to increase flow resistance limited the cooling performance loss to only 0.5 %. As shown in Fig. 15(a), Liu et al. [149] designed a cabinet structure incorporating a pressure chamber and porous panel, with additional baffles placed between servers to prevent fluid bypassing and ensure uniform flow distribution to each server. Studies have found that using porous panels and reducing the gap between servers can significantly improve flow uniformity within the cabinet. Furthermore, for cabinet structures equipped with distribution chambers and porous panels, Ni and Yu [150] introduced a baffle in the middle of the distribution chamber, dividing the cooling container into two sections, each with its own coolant inlet, as illustrated in Fig. 15(b) Their investigation into the effects of various distribution chamber configurations, inlet locations, and porous panel parameters on cabinet cooling performance resulted in more uniform flow distribution. They recommended a porous panel hole radius of 7.5 mm for optimal performance.

Moreover, manifold-based distribution methods can achieve more uniform flow distribution. Saini et al. [151] utilized a distribution manifold to optimize the flow field in a cooling tank, achieving nearly equal volumetric flow rates across each outlet. As shown in Fig. 16, Li et al. [152] proposed a structure combining manifold inflow with porous

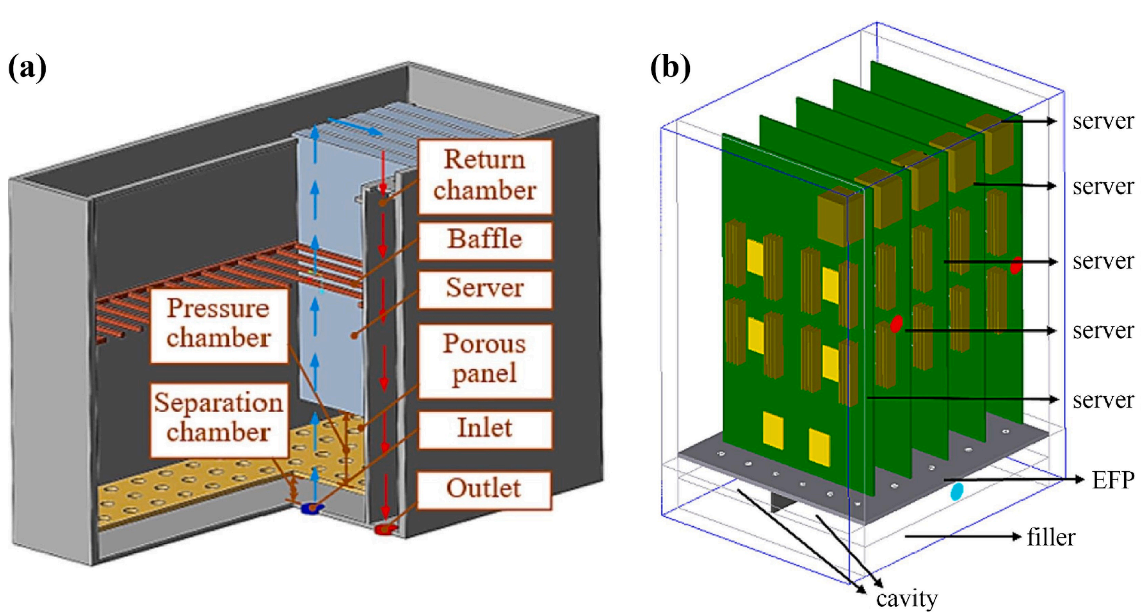


Fig. 15. Schematic of (a) porous panel and server baffles [149], (b)dual inlet porous panel [150].

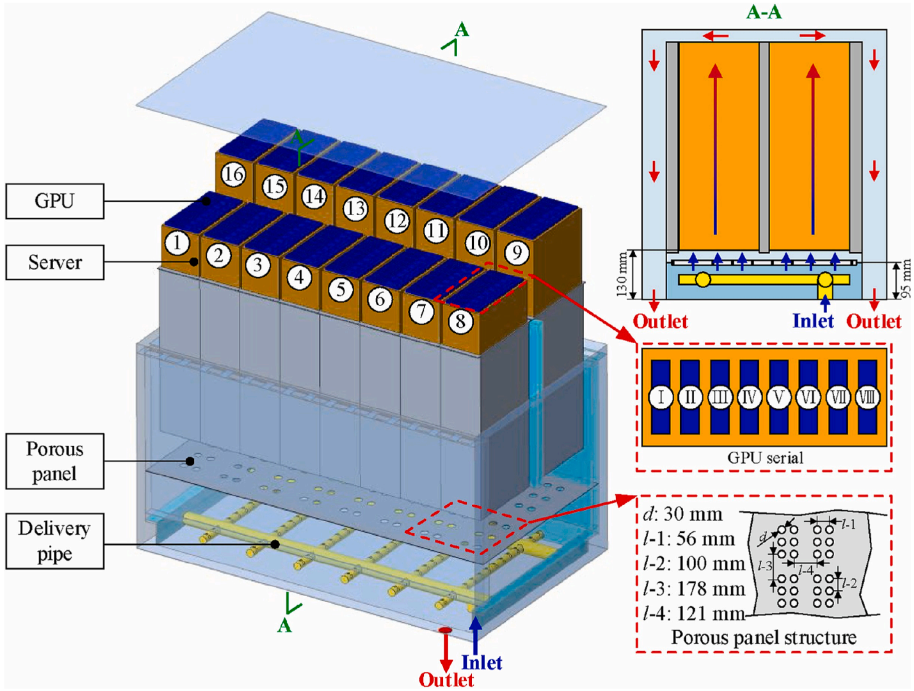


Fig. 16. Manifold and porous panel distribution method [152].

panel distribution. Simulation results demonstrated excellent thermal uniformity, maintaining temperature deviations within 5 °C across 128 GPUs under various operating conditions. Notably, the progression from cabinets without distribution chambers to those combining manifolds and porous panels results in increasing flow distribution uniformity, but also leads to a rise in pressure drop. So, selecting an optimal cabinet design requires a comprehensive evaluation of the thermal load cooling requirements and the trade-offs associated with pump power consumption and implementation costs.

However, this uniform liquid distribution method has exposed the issue of localized hotspots as chip heat flux continues to increase. Specifically, the uniform flow provides excessive cooling for components like memory, while being insufficient for high-heat-flux chips. Therefore, it is necessary to investigate non-uniform flow distribution technologies. These should combine chip-level and server-level heat sinks and flow channel designs to first meet the cooling needs of hotspots and then leverage the advantages of immersion cooling to cool other low-heat components. Currently, there is a lack of research at the cabinet level addressing high-heat flux localized hotspots. Table 3 summarizes current literature on single-phase immersion cooling research for servers and cabinets with higher heat flux densities. It is found that in current single-phase immersion cooling studies, the heat flux of individual servers is generally below 1 kW. Furthermore, the heat flux of chips is typically less than 400 W, with most heat flux densities below 10 W/cm^2. From an application perspective, research on heat dissipation performance within this range may no longer hold significant value.

5. Evaluation of single-phase immersion cooling systems

Understanding the key performance metrics of data centers is crucial for implementing targeted optimizations. A deep comprehension of the characteristics and advantages of single-phase immersion cooling systems aids in identifying and unlocking their potential for further improvement.

5.1. Energy efficiency evaluation metrics

In academic research on thermal management, scholars often focus on the pressure drop of heat sinks and servers or the pump power of the primary coolant loop. However, in practical applications, the cooling of the liquid coolant itself also consumes energy, such as the pump power required for the secondary cooling water loop. Therefore, in the practical implementation of SPIC systems, a more comprehensive energy efficiency metric is necessary to account for the total power consumption of the cooling system.

Power Usage Effectiveness (PUE) is a critical parameter for data centers, defined as the ratio of IT equipment power consumption to total power consumption [158]. The theoretical minimum value of PUE is 1, indicating that all energy is utilized solely by IT equipment. A PUE value closer to 1 reflects better energy-saving performance of the data center [159].

$$PUE = \frac{P_{ICT}}{P_{IT}} = \frac{P_{IT} + P_{\text{cooling}} + P_{\text{other}}}{P_{IT}}. \tag{3}$$

Due to the need for heat exchange with the external environment, cooling energy consumption can vary significantly based on local climate and temperature conditions, which in turn affects the PUE [160]. Data centers in regions with higher average temperatures generally exhibit higher PUE values, making it difficult to compare the energy efficiency of data centers located in different regions fairly [161]. Nonetheless, PUE remains the most widely used metric for evaluating the energy efficiency of data center cooling systems.

Given the complexity of total energy consumption in data centers, a partial Power Usage Effectiveness (*pPUE*) metric has been defined, which focuses on the energy consumption of specific subsystems within the data center infrastructure [162]. This metric can be used to evaluate the relationship between the energy consumption of the cooling system and that of the IT equipment.

$$pPUE = \frac{P_{IT} + P_{\text{cooling}}}{P_{IT}}. \tag{4}$$

It is worth noting that in laboratory studies on SPIC, the *PUE* generally refers solely to the power usage efficiency of the cooling system, namely *pPUE*.

The Energy Saving Ratio (ESR) is another key parameter for evaluating the performance of cooling systems in data centers. It reflects the

Table 3
Studies on single-phase immersion cooling at server and cabinet levels.

Reference	Technical description	Working fluid	Parameter condition	Heat load	Major content or conclusions
[62]	Server-level Forced convection	Fluoride	$V_f = 5.1$LPM $T_f = 20$℃	$Q_{tot} = 1421.5$ W $Q_{max} = 270$ W	Compared to high-viscosity fluorinated liquids, using low-viscosity fluorinated liquids increased the average Nu by 57.3 % while reducing pressure drop by 59 %.
[61]	Server-level Forced convection	Mineral oil, synthetic oil, silicone oil	$V_f = 3.7$LPM $T_f = 30$℃	$Q_{tot} = 860$ W $Q_{max} = 200$ W $q_{max} = 3.8$ W/cm^2	Compared to forced air cooling and cold plate cooling, SPIC reduces the PUE by 20.8 % and 17.6 %, respectively. Viscosity has the greatest impact on SPIC performance.
[53]	Server-level Forced & natural convection	Fluoride	$Re = 16000 \sim 80000$ $T_f = 20 \sim 40$℃	$Q_{tot} = 1332$ W $Q_{max} = 270$ W $q_{max} = 30$ W/cm^2	Compared to natural convection, the coolant mean temperature and PUE in a forced convection system are reduced by 55.5 % and 11.6 %, respectively, achieving a PUE of 1.07.
[114]	Server-level Forced convection	EC110	$V_f = 1.7 \sim 8.3$LPM $T_f = 30 \sim 38$℃	$Q_{tot} = 300$ W $Q_{max} = 120$ W $q_{max} = 8.6$ W/cm^2	Simulation and response surface methodology were employed to optimize seven key parameters in the SPIC system and analyze their interactive effects.
[144]	Server-level Jet-assisted forced convection	FC3283	$V_f = 70$LPM $T_f = 30$℃	$Q_{tot} = 2050$ W $Q_{max} = 375$ W $q_{max} = 15$ W/cm^2	The performance of traditional and jet-impingement-assisted SPIC systems was compared, analyzing the effects of different jet angles and flow rates on the performance of jet-assisted immersion liquid cooling.
[153]	Server-level Forced convection	Mineral oil	$V_f = 0.5 \sim 2.5$LPM $T_f = 25 \sim 50$℃	$Q_{tot} \approx 220$ W $Q_{max} = 95$ W	The experiment observed the effects of changing volumetric flow rate and inlet oil temperature on the thermal performance and power consumption of servers. The PUE ranged from 1.03 to 1.17.
[154]	Server-level Forced convection	Mineral oil, EC-100	$V_f = 0.5 \sim 3.8$ LPM $T_f = 15 \sim 25$ ℃	$Q_{tot} = 500$ W $Q_{max} = 95$ W	Mineral oils with lower viscosity exhibit better cooling performance. As the inlet temperature increases, the density of the dielectric fluid decreases, resulting in a reduction in pumping power.
[145]	Server-level SPIC/WC system	synthetic oil	$V_w = 0.35 \sim 0.85$LPM $T_w = 30 \sim 45$ ℃	$Q_{tot} = 620$ W $Q_{max} = 205$ W	Compared to liquid cooling systems, the designed SPIC/WC cold plate cooling technology reduces power consumption by over 20.7 %.
[155]	Server-level Forced & natural convection	EC-110	$V_w = 0.25$ m/s $T_w = 30$ ℃ $V_f = 18.9$ LPM $T_f = 30 \sim 50$ ℃	$Q_{tot} = 1120 \sim 3760$ W $Q_{max} = 400$ W	Through numerical simulations, the thermal management performance of forced and natural convection SPIC was compared in an 8GPU + 2CPU server under varying conditions, including the number of cold plates, inlet fluid temperature, and server power.
[139]	cabinet level Forced convection 2 servers	Fluoride	$V_f = 2 \sim 8$ LPM $T_f = 18$ ℃	$Q_{tot} = 1750$ W $Q_{max} = 350$ W $q_{max} = 6.67$ W/cm^2	Compared to pro-gravity flow, the chip case temperature is reduced by 33.8 % and the PUE is reduced by 1.4 % under anti-gravity flow conditions.
[95]	cabinet level Forced convection 12U cabinet	PAO-2	$V_f = 6$ m^3/h $T_f = 38$ ℃	$Q_{tot} = 13 \sim 17$ kW $Q_{max} = 330$ W	A scalable modular immersion cooling system was proposed, and its cooling capacity was validated through simulations and experimental verification.
[156]	cabinet level Forced convection 48U cabinet	Fluoride	$V_f = 6$ m^3/h $T_f = 38$ ℃	$Q_{tot} = 60$ kW $Q_{max} = 350$ W	SPIC system designed for commercial data centers was studied, focusing on the role of filler modules, flow field uniformity design, and optimized heat sink design.
[149]	cabinet level Forced convection 10–40 servers	Synthetic oil	$V_f = 1 \sim 10$ m^3/h $T_f = 40$℃	Simplified model	The use of porous panels can improve flow uniformity within the tank. Orthogonal analysis and the entropy weight method were applied to evaluate four key parameters.
[52]	cabinet level Natural convection 32 servers	Silicon oil soybean oil, FC3282 FC43	$T_w = 15 \sim 35$℃	$Q_{tot} = 440$ W × 32 $Q_{max} = 140$ W	The proposed solution improves space utilization and system stability, with the system's PUE reduced to below 1.04. Fluoride with low viscosity is more suitable for natural convection cooling.
[152]	cabinet level Forced convection 16 servers	BINGYI 797	$V_f = 0.13 \sim 2.19$ m^3/h $T_f = 25$ ℃	$Q_{tot} = 2800$ W × 16 Each server GPU: 350 W × 6	The manifold and porous panel distribution structure can achieve good flow distribution, improving temperature uniformity.
[98]	cabinet level Forced convection 54 servers	Mineral oil	$V_f = 7.5$ m^3/h $T_f = 40$ ℃	$Q_{tot} = 21.6$ kW	The study investigates the system's response to static and dynamic workloads, as well as the impact of sudden changes.
[157]	cabinet level Forced convection 60 servers	Fluoride	Each server: 3 ~ 7LPM $T_f = 40$ ℃	$Q_{tot} = 1$ kW × 60 $Q_{max} = 230$ W	The reliability of various heat sinks and thermal interface materials was validated. It was found that under immersion conditions, signal integrity meets the required standards.

energy-saving effectiveness of an optimized cooling system. ESR is defined as the proportion of electrical energy saved by the optimized cooling system relative to the total energy consumption of the original cooling system and can be expressed as follows [163]:

$$ESR = \frac{C - A}{C} \times 100\%. \tag{5}$$

where C represents the total energy consumption of the original cooling system, and A represents the total energy consumption of the optimized cooling system.

Data centers using SPIC typically have higher coolant outlet temperatures, offering significant potential for heat recovery. The Energy Reuse Factor (ERF) can be used to measure the performance of heat recovery in data centers [164]. It is defined as the ratio of energy reused by the data center to the energy input into the data center. Energy Reuse Effectiveness (ERE) represents the overall Power Usage Effectiveness (PUE) of the data center, taking into account the energy that is reused [165]. If there is no energy reuse (ERF = 0), the ERE equals PUE. However, when energy is reused (ERF > 0), the ERE will be lower than the PUE.

$$ERF = \frac{P_{reuse}}{P_{ICT}}. \tag{6}$$

$$ERE = \frac{P_{ICT} - P_{reuse}}{P_{IT}} = (1 - ERF)PUE. \tag{7}$$

where P_{reuse} represents the power of energy reused by the data center. In addition to these metrics, some researchers have employed methods such as systems cost-effectiveness analysis to gain a deeper understanding of the thermal and fluid dynamics of various components in immersion cooling systems [42]. A detailed review of data center cooling system metrics has been conducted by Shao et al. [166].

5.2. Energy efficiency advantages

In terms of cooling system energy consumption, current air cooling methods exhibit an average PUE of approximately 1.5–1.6 and an average ESR ranging from 35 % to 40 %. Liquid cooling methods offer improved efficiency, with average PUE and ESR values of 1.1–1.2 and 45 %–50 %, respectively [167]. Immersion cooling demonstrates even lower PUE levels. Under laboratory conditions, SPIC systems typically achieve a PUE below 1.1 [53,168], with some studies reporting PUE values lower than 1.04 [52,153] In comparison, even direct fresh air cooling data centers reach a PUE of only 1.08 [169]. To further enhance power utilization efficiency and enable multi-gradient energy utilization, researchers have effectively combined immersion cooling with waste heat recovery and reuse systems. These integrated systems have demonstrated the potential to reduce ERF, lower operational costs, and decrease carbon emissions [170,171].

On the commercial front, SPIC technology has received significant attention in data centers, and practical applications have demonstrated that SPIC-based data centers can substantially reduce operational energy costs. In 2018, Inspur introduced the TS4220LC supercomputing platform employing immersion liquid cooling, achieving a PUE of 1.1–1.2 [167]. Green revolution cooling (GRC) designed an immersion cooling system with a single rack cooling capacity of up to 200 kW, delivering a PUE of 1.03 and an ESR of 95 % [172]. Alibaba implemented a pump-driven SPIC system using S5X as the working fluid, achieving a PUE of 1.07 [173]. Similarly, Baidu, in collaboration with Intel, developed SPIC cabinets capable of dissipating over 36 kW with a PUE below 1.1 [174]. Table 4 summarizes the energy efficiency metrics of single-phase immersion data centers and related products at a commercial level. It can be observed that current commercial applications of single-phase immersion cooling in data centers have achieved a PUE below 1.1, with immersion cabinet cooling capacities reaching up to 200 kW.

Due to the significant application potential of SPIC systems in data centers, researchers have increasingly begun exploring system-level studies, including secondary-side cooling water loops for SPIC in data centers. Liu et al. [178] conducted experimental investigations on SPIC systems, examining their cooling capacity and energy consumption. Their findings revealed that both coolant flow rate and cooling water flow rate significantly impact the heat transfer coefficient and energy consumption of SPIC systems. They also derived heat transfer correlations and pressure drop equations with high predictive accuracy. Lionello et al. [179] developed a universal model for simulating energy and mass transfer processes in immersion cooling systems. This model demonstrated good agreement with experimental data and can be used to accelerate the design cycle of complex SPIC systems. Wang et al. [180] investigated the effects of operating temperatures of secondary-side water-cooled cooling towers and system operation strategies on SPIC data centers. Their study revealed that ambient temperature significantly impacts energy consumption, and the adoption of effective system operation strategies can markedly improve operational efficiency. Liu et al. [181] further designed a secondary side cooling system based on liquid air, tailored for immersion liquid-cooled data centers. Modeling studies show that compared to immersion cooling data centers with secondary side using evaporative cooling towers, the secondary side with a liquid air cooling system achieves a 5 % reduction in pPUE. The cooling system reaching a pPUE of 1.006. Overall, current research on system-level SPIC applications remains limited. To further improve the performance of SPIC systems in data centers, more research at the system level is necessary in the future.

Table 4
Energy efficiency metrics of SPIC at commercial level.

Company	Condition	Model	PUE	ESR
Tokyo Institute of Technology and GRC[175]	IT power per rack: 28.9 kW	TSUBAME-KFC	1.09	/
Inspur[167]	/	TS4220LC	1.1–1.2	30 %
GRC[172]	IT power per rack: 200 kW	Immersion Cooling System ICEraQ Server	1.03	95 %
GRC[176]	IT power per rack: 50 kW	Immersion Cooling System ICEtank Server	1.03–1.09	90 %
Alibaba[173]	Total IT power: 2000 kW	/	1.07	36 %
Baidu and Intel [174]	IT power per rack: over 36 kW	Scorpio 5.0 immersion cooling rack solution (ICRS)	< 1.1	/
Sustainable Metal Cloud[177]	IT power per rack: over 70 kW	HyperCube	1.03–1.1	> 40 %

6. Conclusion and prospect

6.1. Conclusion

This study presents a comprehensive review of single-phase immersion cooling in data centers, with a focus on the performance of immersion coolants, chip-level heat dissipation enhancement strategies, structural optimization of servers and cabinets, and the advantages of cooling systems. Several important conclusions have been drawn.

(1) The advantage of oil-based coolants lies in their high heat transfer coefficient and large specific heat capacity. Fluorinated liquids, on the other hand, have the benefits of low viscosity, high compatibility, and stability, but they have a lower thermal conductivity and are more expensive. Given the material compatibility of oil coolants, it is necessary to establish a compatibility report for various main materials when using new oils. High-frequency, high-bandwidth signal servers should undergo specific optimization to adapt to the immersion environment.

(2) Single-phase immersion cooling should not be limited to addressing the issue of local hotspots. The use of heat pipes or heat spreaders, as well as adopting active convective heat transfer methods such as direct-to-chip cooling, has the potential to cool 1 kW chips. At the same time, expanding the effective heat dissipation area and increasing the convective heat transfer coefficient can achieve the heat dissipation limit for a single chip of over 1 kW. However, research on this aspect is severely lacking, which limits the application scenarios of single-phase immersion cooling.

(3) Reasonable coolant flow channels need to be implemented within single-phase immersion cooling servers in data centers to enhance the cooling performance of key electronic components while reducing pressure losses. Methods such as baffles, localized turbulence enhancement, and flow redistribution technology are relatively effective. At the cabinet level, more uniform flow distribution technology typically results in increased pressure loss and costs. However, in many cases, due to the uneven heat distribution inside servers, overly uniform flow distribution is not necessary. Localized flow distribution techniques targeting

hotspots are more beneficial for highlighting the advantages of immersion cooling.

(4) Single-phase immersion cooling systems do not require chillers or air conditioning units, which can significantly reduce the footprint of the cooling system. By eliminating cooling fans, these systems operate with low noise and energy consumption, with a typical power usage effectiveness of less than 1.1, leading to a significant reduction in the energy costs of data center operations.

6.2. Prospect

In recent years, with the increasing attention on single-phase immersion cooling, a large number of research results have been produced. However, existing systems still have many shortcomings, which define future research opportunities. These opportunities include: standardizing the selection of cooling fluids and adopting advanced technologies in the design of cooling fluids, enhancing the heat dissipation capacity of high-heat-flux chips, designing server structures with multiple high-heat-flux chips, and innovatively optimizing single-phase immersion cooling systems for data centers.

(1) To date, there is no universally accepted comparative standard for the selection of immersion cooling liquids. Research into selection criteria for cooling fluids under various general cooling conditions is essential for ensuring the operational stability of data centers, reducing energy consumption, and achieving standardized cooling fluid design. This comparative standard should focus on key factors such as thermal conductivity, electrical insulation, chemical stability, and environmental friendliness. In the design of cooling fluids, it is necessary to integrate advanced methods such as AI and big data to further develop molecular structure designs for cooling liquids. Based on comparative selection criteria, more efficient, environmentally friendly, and cost-effective cooling fluids can be designed.

(2) As illustrated in Fig. 17(a), under high heat flux conditions of 100 W/cm^2, achieving a temperature difference of 25 °C without expanding the effective surface area would require a heat transfer coefficient of 4 × 10^4 W/(m•K), which is challenging for single-phase immersion cooling systems. Fig. 17(b) shows that increasing the effective heat dissipation area significantly reduces the required average convective heat transfer coefficient. By coupling this approach with convection enhancement techniques, as depicted in Fig. 17(c), efficient thermal management for high-heat-flux chips can be achieved. Thus, to avoid excessive pump power due to high coolant flow rates, it is recommended to implement both surface area expansion and convection enhancement methods for cooling high-heat-flux chips. Area expansion can be achieved through advanced techniques such as heat pipes, vapor chamber, and 3D thermosiphon [7,182] et al. Convection heat transfer can be strengthened using forced convection in limited spaces, jet impingement, or other turbulence-inducing methods. To ensure flow velocity, it is typically recommended to position the cooling liquid inlet near the chip heat sink. Moreover, there is currently limited research on using SPIC methods to cool high-heat-flux 3D stacked chips. Embedded microchannel technology offers significant advantages in 3D chip thermal management, while immersion cooling fluids, with their excellent material compatibility and electrical insulation properties, can meet the requirements of embedded cooling systems. Additionally, the immersion environment provides efficient heat dissipation performance. Combining direct-to-chip microchannels with natural convection or pump-to-server solutions could be a promising technological pathway. Achieving the cooling of higher heat flux chips and 3D chips will significantly expand the applicability of SPIC data centers.

(3) The demand for AI computing power is growing exponentially, with high-performance 8-GPU servers typically having a heat load of over 5 kW, and the highest models like NVIDIA's Blackwell B200 servers even exceeding 8 kW [183]. As shown in Fig. 18, the schematic diagrams illustrate the use of cold-plate water cooling and single-phase immersion cooling for multi-heat-source, high-heat-flux servers. Cold-plate water-cooled servers exhibit a complex piping structure and require additional fans to cool smaller heat-generating components inside the server. Although SPIC requires larger heat sinks, it eliminates the need for fans and complex piping, significantly reducing maintenance demands and energy consumption. Single-phase immersion cooling systems, due to their high heat capacity, large flow rates, and direct-contact cooling mode, have inherent advantages in cooling such multi-source, high-heat flux servers. However, most of the current research has mostly focused on servers with heat loads under 5 kW and fewer than four major chip heat sources, failing to fully leverage the potential of

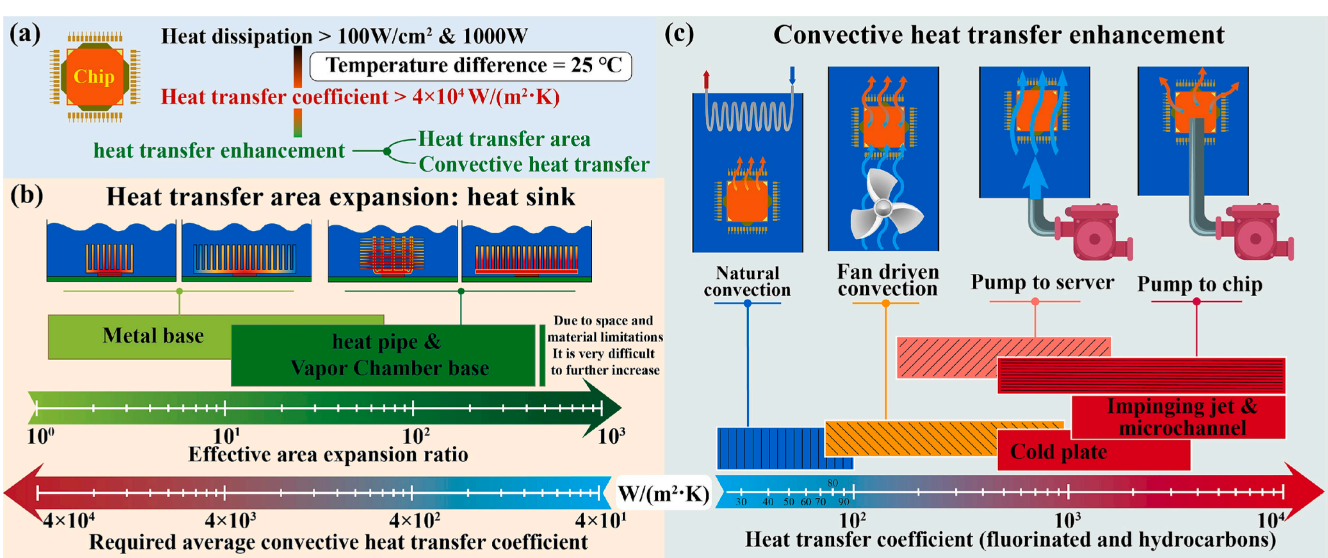


Fig. 17. Single-phase immersion cooling at the chip level: (a) heat dissipation requirements for chip, (b) effect of increasing the effective heat dissipation area, (c) convective heat transfer coefficients of various SPIC methods.

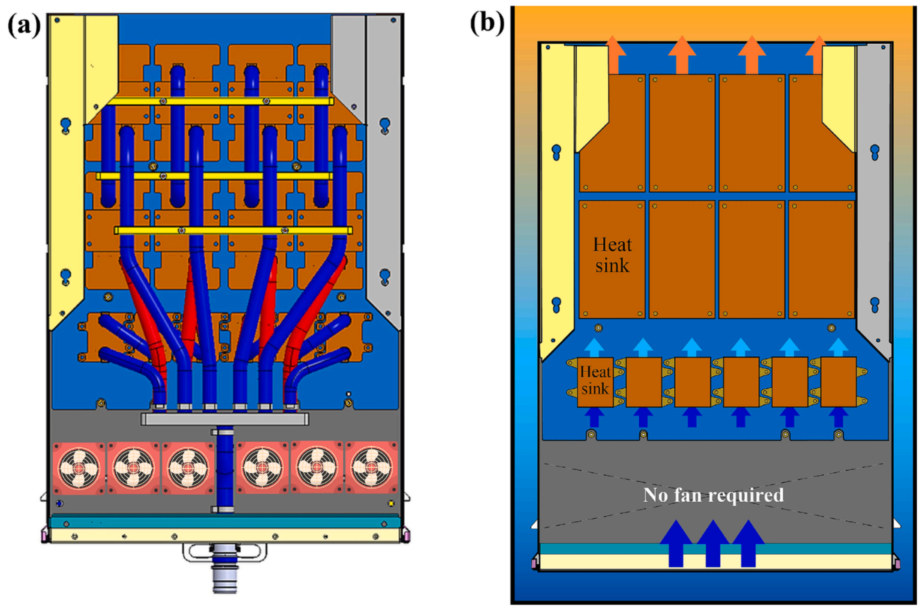


Fig. 18. Cooling methods for multi heat source AI servers: (a) cold plate water cooling [17], (b) single-phase immersion cooling.

immersion cooling systems' high heat capacity and flow rates. By combining chip-level heat dissipation enhancement methods, developing specialized cooling solutions for high-heat flux AI servers could open a new research direction for the sustainable and energy-efficient development of data centers.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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